

Toh chalo shuru karte hain ek independent chapter, fluid mechanics... Isme thoda sa basic mechanics ka use hoga... Exam ke POV (Point of View) se JEE Mains mein isse direct formula base, basic concept par kaafi question pooche gaye hain aur ADV (Advanced) ke POV se chapter mixing type kaafi question pooche gaye hain... Bas ye yaad rakhna, kisi bhi question se darna nahi hai... Toh shuru karein phir...

YOUR EXPRESSIONS



When the teacher announces a new chapter



Ideal Fluid

- ★ No shear stress
- ★ There is no friction between any two layers of fluid (Non Viscous)
- ★ Density same $\rightarrow$ uniform
- ★ Incompressible
- ★ Irrotational



Rate of aane wali = Rate of jane wali

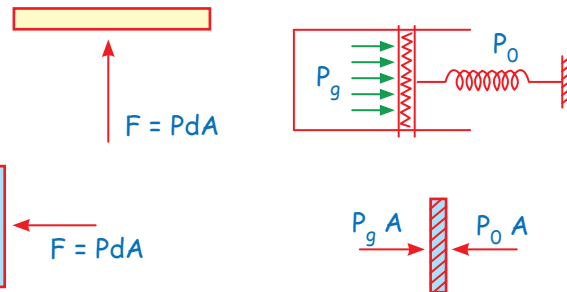
Pressure

- ★ It is force acting normal to the surface per unit area.

$$P = \lim_{\Delta A \rightarrow 0} \left(\frac{F}{\Delta A} \right) \text{ (Point function)}$$

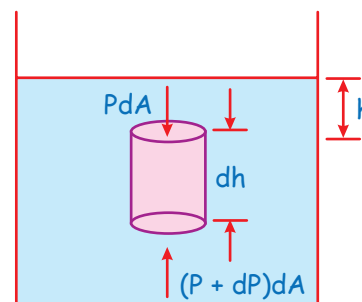
- ★ Direction of force due to pressure will always be perpendicular to the surface

$$\text{Force} = F = PdA$$



Variation of Pressure with Depth in Liquid

Consider a small cylinder of cross section dA and length dh made up of water at a depth h



This element of fluid will be at rest

$$\therefore F_{\text{net}} = 0$$

$$\Rightarrow PdA + \rho g dA dh - (P + dP) dA = 0$$

$$\Rightarrow dP = \rho g dh$$

$$\int_{P_0}^P dP = \int_0^h \rho g dh$$

$$P = P_0 + \rho gh$$

Seedhi baat, if fluid is at rest, toh neeche jaane par pressure ρgh ke hisaab se badhega.

$$P = P_0 + \rho gh$$

$$P - P_0 = \text{gauge pressure}$$

Haan haan, mujhe malum hai tumne yeh chemistry mein bhi padh rakha hai. Main toh bas yun hi bata raha tha bhai, chapter toh yahin se shuru hota hai.

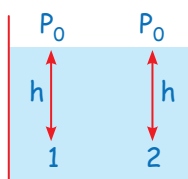


Pressure Variation Along Horizontal Line

$$1. P_1 = P_0 + \rho gh$$

$$P_2 = P_0 + \rho gh$$

$$P_1 = P_2$$



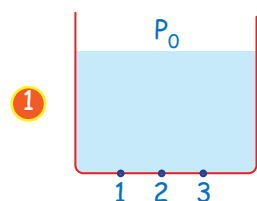
SKC

Agar P_0 ko zero maan loon ya neglect kar doon, toh kisi jagah par jo Pressure aayega woh uss jagah ka Gauge Pressure hoga.

Bhai chalo sabse pehle fluid static mein kisi point par pressure nikaalna seekhte hain... solve each and every fundamental questions

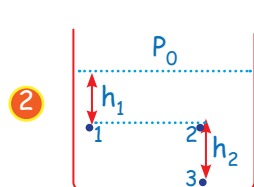
Pressure Variation Along Vertical Line

Q. Write the pressure at given points in following question.



$$P_1 = P_0 + \rho gh$$

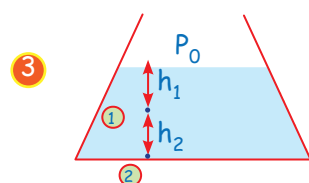
$$P_1 = P_2 = P_3$$



$$P_1 = P_0 + \rho gh_1$$

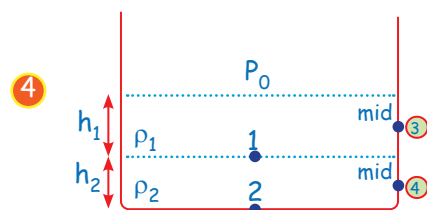
$$P_2 = P_0 + \rho gh_1$$

$$P_3 = P_0 + \rho gh_1 + 0 + \rho gh_2$$



$$P_1 = P_0 + \rho gh_1$$

$$P_2 = P_1 + \rho gh_2$$



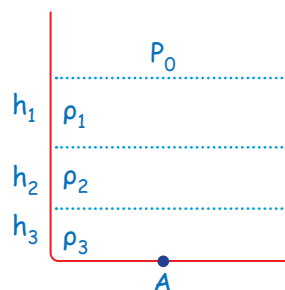
$$P_1 = P_0 + \rho_1 gh_1$$

$$P_2 = P_0 + \rho_1 gh_1 + \rho_2 gh_2$$

$$P_3 = P_0 + \rho_1 g h_1 / 2$$

$$P_4 = P_0 + \rho_1 gh_1 / 2 + 0 + \rho_2 g h_2 / 2$$

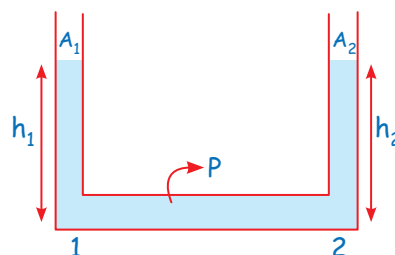
5



$$P_A = P_0 + \rho_1 gh_1 + \rho_2 gh_2 + \rho_3 gh_3$$

Question on U Shape Tube

Q. Find P_1 & P_2 and relationship between h_1 and h_2 .



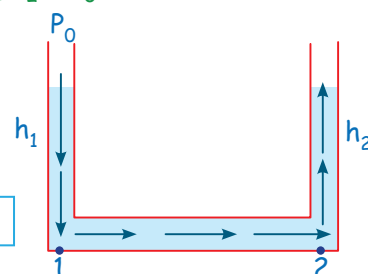
$$\text{Sol. } P_0 + \rho gh_1 + 0 - \rho gh_2 = P_0$$

$$h_1 = h_2$$

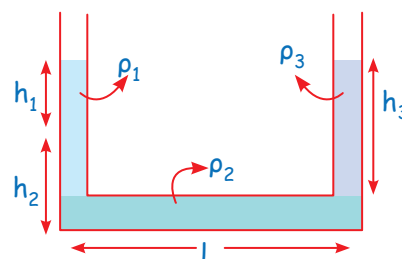
$$P_1 = P_0 + \rho gh_1$$

$$P_2 = P_0 + \rho gh_2$$

$$P_1 = P_2 \rightarrow h_1 = h_2$$

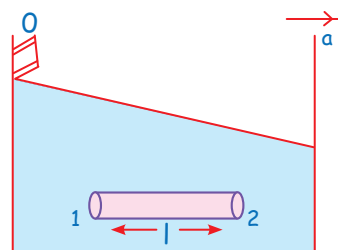


Q. Write pressure equation.



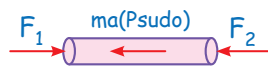
$$\text{Sol. } P_0 + \rho_1 gh_1 + \rho_2 gh_2 + 0 - \rho_3 gh_3 = P_0$$

Pressure Variation Along Horizontal Direction if Beaker is Accelerated.



Consider a hypothetical horizontal cylinder of length l and cross-section area A and same density as fluid as shown in figure.

FBD wrt beaker



$$F_1 = F_2 + ma$$

$$P_1 A = P_2 A + \rho A l a$$

$$P_1 - P_2 = \rho a l$$

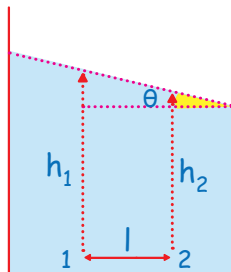
$$P_1 = P_0 + \rho g h_1$$

$$P_2 = P_0 + \rho g h_2$$

$$P_1 - P_2 = \rho g (h_1 - h_2) = \rho a l$$

$$\tan \theta = \frac{h_1 - h_2}{l} = \frac{a}{g}$$

$$\Rightarrow \tan \theta = a/g$$



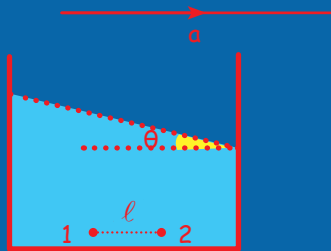
SKC

Dekh bhai, oopar wala derivation chahe toh bhool ja, lekin result yaad kar le, aur woh yeh hai ki agar beaker horizontally acc. (accelerated) hai, toh ab same level par pressure same nahi hoga. Jaise neeche jaane par pressure $\rho g h$ ke hisaab se badhta tha, waise hi ab pseudo force ki taraf pressure $\rho a l$ ke hisaab se badhega.

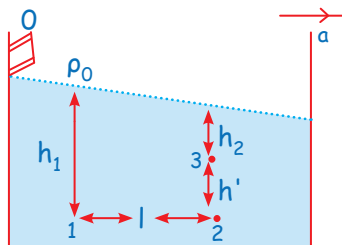
$$P_1 \neq P_2$$

$$P_2 + \rho a l = P_1$$

$$\tan \theta = a/g$$



Q. Find pressure at point 1, 2 and 3.

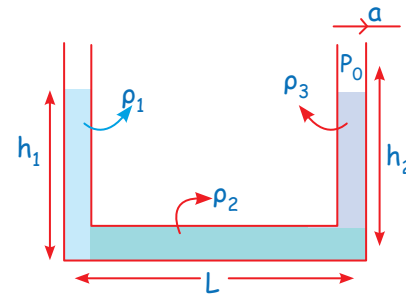


$$\text{Sol. } P_1 = \rho_0 + \rho g h_1$$

$$P_2 = \rho_0 + \rho g h_1 - \rho a l$$

$$P_3 = \rho_0 + \rho g h_1 - \rho a l - \rho g h'$$

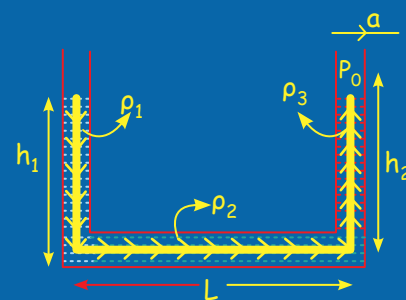
Q. Write the pressure equation for the given diagram



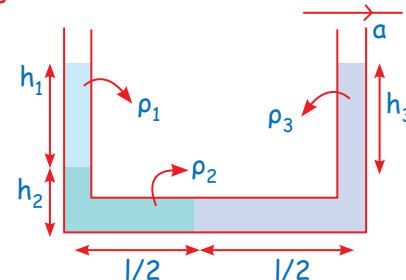
$$\text{Sol. } P_0 + \rho_1 g h_1 + \rho_2 a l - \rho_3 g h_2 = P_0$$

SKC

Isse hum bolenge "dubki method," bole toh ek side se dubki maaro aur swim karte hue doosri side nikal jaao aur pressure ki eqn (equation) likhte jaao

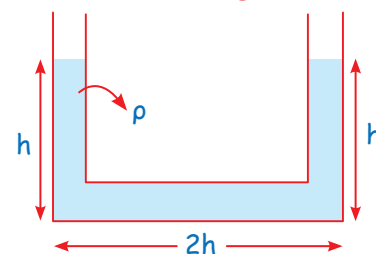


Q. Write the pressure equation for the given diagram.

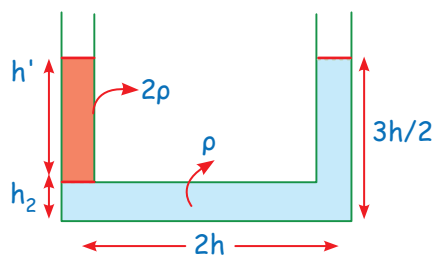


$$\text{Sol. } P_0 + \rho_1 g h_1 + \rho_2 g h_2 - \rho_2 a l / 2 - \rho_3 a l / 2 - \rho_3 g h_3 = P_0$$

Q. In following youtube what vol of liq of density 2ρ should be filled in left arm so that 50% of liq in left arm shifted to right arm.



Sol.

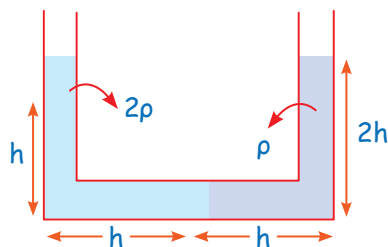


$$P_0 + 2\rho gh' + \rho gh/2 + 0 - \rho g 3h/2 = P_0$$

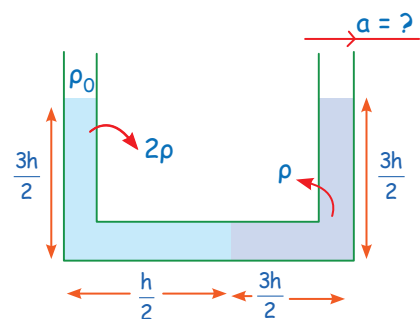
$$2h' - h = 0 \quad \boxed{h' = h/2}$$

$$\text{Ans.} = \frac{Ah}{2} \quad A \rightarrow \text{Area of cross section}$$

Q. What should be acceleration of beaker in horizontal line (+x axis) so that 75% of horizontal part of U-tube is filled by liquid ρ ?



Sol.



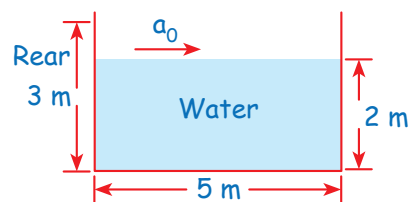
$$P_0 + 2\rho g \frac{3h}{2} - 2\rho a \frac{h}{2} - 0 - \rho a \frac{3h}{2} - \rho g \frac{3h}{2} = P_0$$

$$\boxed{a = 3g/5}$$

Q. An open rectangular tank 5 m × 4 m × 3 m containing water upto a height of 2 m is accelerated horizontally along the longer side as shown in figure below.

- Determine the maximum acceleration that can be given without spilling the water.
- Calculate the percentage of water spill over, if this acceleration is increased by 20%.
- If initially, the tank is closed at the top and is accelerated horizontally by 9 m/s^2 , find the gauge pressure at the bottom of the front wall of the tank.

(Take $g = 10 \text{ m/s}^2$).



Sol. (a) Volume of water inside the tank remains constant

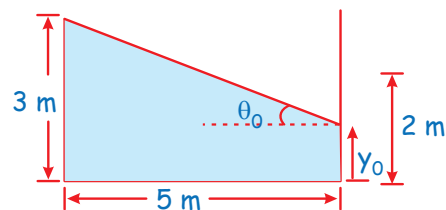
$$\frac{3+y_0}{2} \times 5 \times 4 = 5 \times 2 \times 4$$

$$\text{or } y_0 = 1 \text{ m}$$

$$\therefore \tan \theta_0 = \frac{3-1}{5} = 0.4$$

$$\text{since, } \tan \theta_0 = \frac{a_0}{g},$$

$$\text{therefore } a_0 = 0.4 g = 4 \text{ m/s}^2$$



(b) When acceleration is increased by 20%

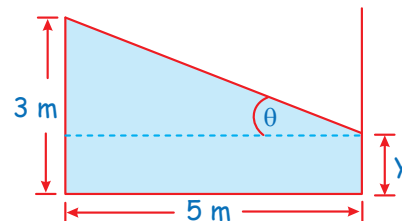
$$a = 1.2 a_0 = 0.48 g$$

$$\therefore \tan \theta = \frac{a}{g} = 0.48$$

Now, $y = 3 - 5 \tan \theta = 3 - 5 (0.48) = 0.6 \text{ m}$
fraction that spills over

$$= \frac{4 \times 2 \times 5 - \frac{(3+0.6)}{2} \times 5 \times 4}{2 \times 5 \times 4} = 0.1$$

percentage of water spill over = 10%

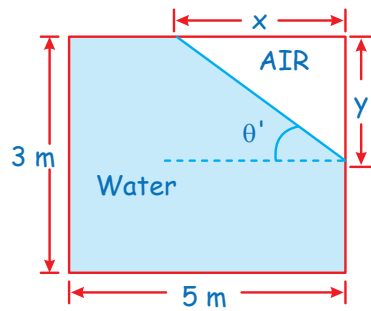


(c) $a' = 0.9 g$

$$\tan \theta' = \frac{a'}{g} = 0.9$$

volume of air remains constant

$$4 \times \frac{1}{2} yx = (5) (1) \times 4$$



since $y = x \tan \theta'$

$$\therefore \frac{1}{2}x^2 \tan \theta' = 5$$

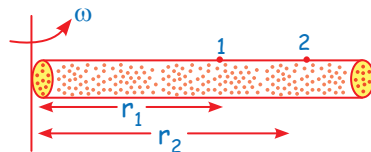
or $x = 3.33 \text{ m}$;

$y = 3.0 \text{ m}$

(front wall contact with air)

Gauge pressure at the bottom of the front wall $p_f = \text{zero}$

ROTATION OF TUBE



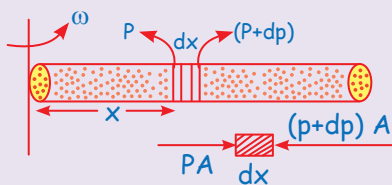
★ $P_2 > P_1$

★ $P_2 - P_1 = \frac{1}{2} \rho \omega^2 (r_2^2 - r_1^2)$

★ Axis se dur jane par Pressure ↑
Yeh kabhi mat bhoolna, acche se yaad kar lo



Proof



$$(P + dP)A - PA = (dm) x \omega^2$$

$$Adp = \rho A dx \cdot x \cdot \omega^2$$

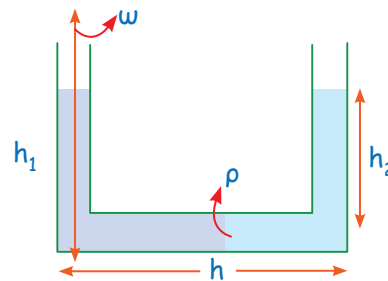
$$\int_{P_1}^{P_2} dp = \rho \omega^2 \int_{r_1}^{r_2} x dx$$

$$P_2 - P_1 = \frac{1}{2} \rho \omega^2 (r_2^2 - r_1^2)$$

r_1 & r_2 are axis from distance



Q. Find $h_2 - h_1$ or difference in height of liq in U tube.

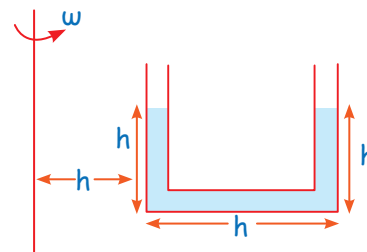


Sol. $P_0 + \rho g h_1 + \frac{1}{2} \rho \omega^2 (h^2 - 0^2) - \rho g h_2 = P_0$

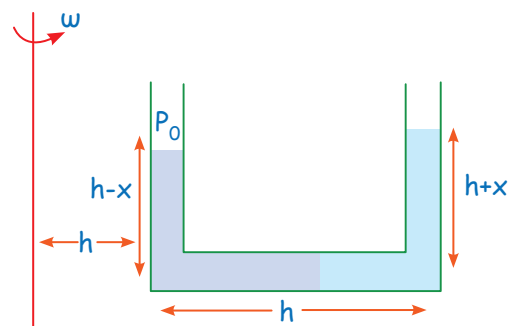
$$g h_1 + \frac{1}{2} \omega^2 h^2 - g h_2 = 0$$

$$(h_2 - h_1) = \frac{h^2 \omega^2}{2g}$$

Q. Find difference in height of liq in both vertical arm after tube is rotated by $\omega = \sqrt{\frac{g}{4h}}$



Sol.

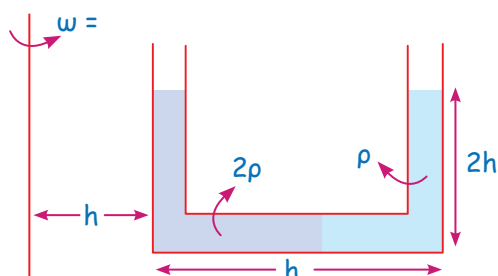


$$P_0 + \rho g (h - x) + \frac{1}{2} \rho \omega^2 [(2h)^2 - h^2] - \rho g (h + x) = P_0$$

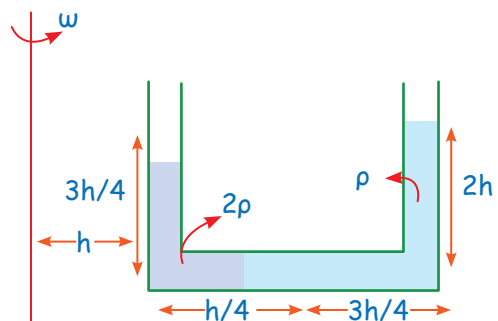
$$\Rightarrow x = \frac{3h}{16} \text{ (After solving)}$$

$$\text{Difference in height } (h + x) - (h - x) = \boxed{\frac{3h}{8}}$$

Q. Find ω so that 75% of base of U tube is filled by liq of density ρ .



Sol.

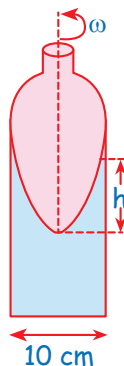


$$P_0 + 2\rho g \frac{3h}{4} + \frac{1}{2} 2\rho \omega^2 \left[\left(\frac{5h}{4} \right)^2 - (h)^2 \right] + \frac{1}{2} \rho \omega^2 [(2h)^2 - (5h/4)^2] - \rho g \frac{9h}{4} = 0$$

Now you can solve. (be careful in calculation)

Q. JEE Mains Favourite question

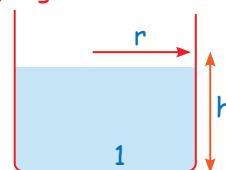
A cylindrical vessel containing a liquid is rotated about the axis so that the liquid rises at its sides as shown in the figure. The radius of vessel is 5 cm and the angular speed of rotation is ω rad/sec. The difference in height h (in cm) of liq at the center of vessel and at the side will be



Sol. $P_0 + \frac{1}{2} \rho \omega^2 (r^2 - 0^2) - \rho gh = P_0$

$$h = \frac{\omega^2 r^2}{2g}$$

Q. Find the force applied by base of the beaker to the liquid (Neglect Atm Pressure)



Pressure at bottom of beaker $P_1 = \rho gh$

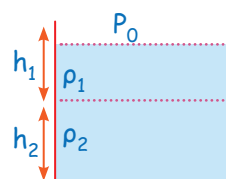
Force applied by liq on to the base of beaker

Sol. $F = P \times \text{Area}$

$$F = \rho gh \cdot \pi r^2$$

$$F = \rho \pi r^2 hg$$

Q. Find force applied by liquid on base of beaker



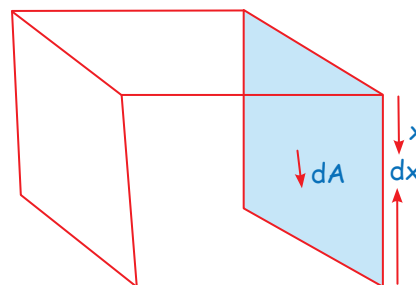
Sol. $(P_0 + \rho_1 gh_1 + \rho_2 gh_2) \times (\text{Area})$

SKC

Agar Pressure Uniform hai to us Pressure ki wajah se lagne wale F ki value, $F = PA$

Pressure
Area

Q. A cube of side length L is completely filled with water of density ρ . (Neglect Atm pressure $P_0 = 0$)



1. Find the force applied by water on base of cube

Ans. $(\rho g L) \times L^2$

2. Find the force applied by water on vertical wall.

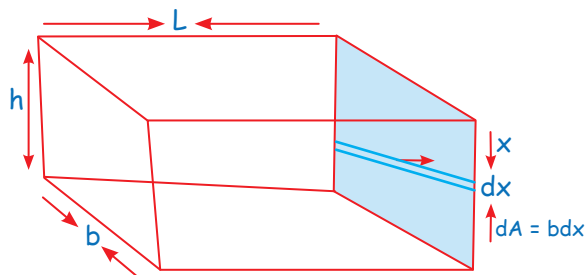
Ans. Force acting on small area dA is,

$$dF = PdA = (\rho gx) \times (Ldx)$$

$$\int dF = \int_0^L \rho g L x dx$$

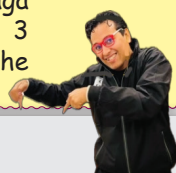
$$F = \rho g L (L^2/2)$$

Q. If this cube is replaced by cuboid (L, b, h), then find the force exerted by liquid on a vertical wall.



Sol. $dF = p dA$
 $\int dF = \int_0^h \rho g x \times b \, dx$
 $F = \rho g b \frac{h^2}{2}$
 $F = 1/2 \rho g h \times b h$

Iss tarah ke sawalon ke liye aap bina integration ke direct yeh shortcut laga kar bhi kar sakte hain. Main neeche 3 example attach kar raha hoon, inhe acche se karo



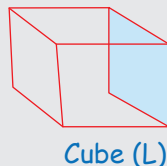
SHORTCUT

Force on one side vertical wall by liquid = (pressure at centre of wall) $\times$ (Area of wall)

$$F = [P_0 + \rho g L/2] \times L^2$$

If $P_0 = 0$ (No atm pressure)

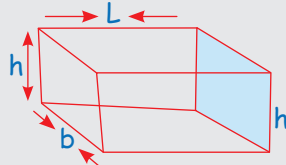
$$F = \rho g L/2 \times L^2$$



$$F = (P_0 + \rho g h/2) \times b h$$

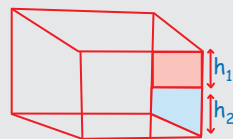
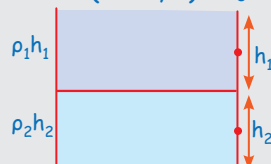
If $P_0 = 0$

$$F = \rho g h/2 \times b h$$



Force by liquid on single side vertical wall

(cube, L) P_0



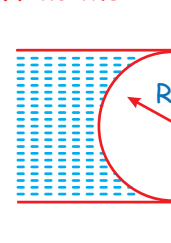
$$F_1 = (P_0 + \frac{\rho}{2} g h_1) \times L h_1 + (P_0 + \rho_1 g h_1 + \rho_2 g \frac{h_2}{2}) \times L h_2$$

upar bala liq ne
lgaya force

niche bala liq ne
lgaya force

Q. A hemisphere of radius R is just submerged is just sinking in water of density ρ . Find the

- Horizontal thrust.
- Vertical thrust.
- Total hydrostatic force.
- Angle of orientation of total hydrostatic force acting on the hemisphere



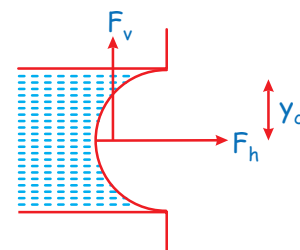
Do not count atmospheric pressure

Sol. (a) Let the horizontal and vertical thrusts on the hemisphere be F_h and F_v respectively

We know that

$$\text{Where } F_h = \rho g y_c A_v \text{ and } y_c = R$$

$$A_v = \pi R^2 \text{ then } F_h = \rho g \pi R^3$$



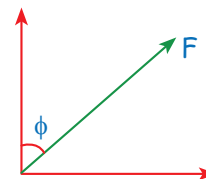
$$(b) V = \text{volume of the hemisphere} = \frac{2}{3} \pi R^3$$

$$\text{we have } F_v = \frac{2}{3} \rho g \pi R^3 \text{ (up)}$$

(c) Hence the net hydrostatic force on the hemisphere is

$$F = \sqrt{F_h^2 + F_v^2}$$

$$F = \sqrt{(\rho g \pi R^3)^2 + \left(\frac{2}{3} \rho g \pi R^3\right)^2} = \frac{\sqrt{13}}{3} \rho g \pi R^3$$



(d) The angle of orientation of the force F is

$$\phi = \tan^{-1} \frac{F_h}{F_v}$$

$$= \tan^{-1} \frac{\rho g \pi R^3}{\frac{2}{3} \rho g \pi R^3} = \tan^{-1} \frac{3}{2}$$

Variation of Pressure with Height

Let ρ_0 be the density of air at the earth's surface where the pressure is atmospheric p_0 , then $\rho_0 = k p_0$

$$p = p_0 e^{\frac{-\rho_0 g h}{p_0}}$$

where p is the pressure at a height $y = h$ above the earth's surface.

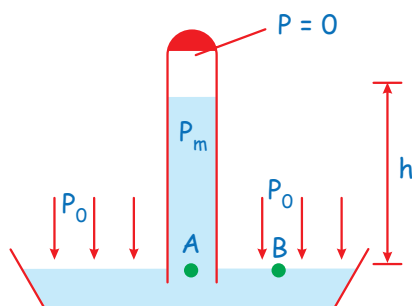
Mercury Barometer

It is a straight glass tube (closed at one end) completely filled with mercury and inserted into a dish which is also filled with mercury as shown in the figure.

Pressure at points A and B are equal and thus

$$\Rightarrow P_B = P_A \Rightarrow p_0 = 0 + \rho g h$$

Hence, $p_0 = (13.6 \times 10^3)(9.81)(0.76) = 1.01 \times 10^5 \text{ Nm}^{-2}$ (or Pa)



PASCAL'S LAW

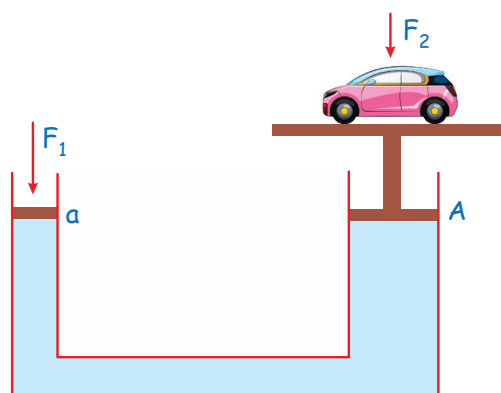
- ★ A pressure applied to a confined fluid at rest is transmitted equally undiminished to every part of the fluid and the walls of the container.
- ★ This principle is used in a hydraulic jack or lift, as shown in the figure.

Hydraulic Jack (Lift)

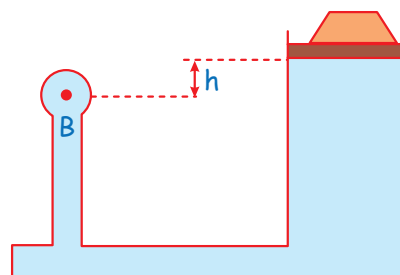
- ★ The pressure due to a small force F_1 applied to a piston of area a is transmitted to the larger piston of area A . The pressure at the two pistons is the same because they are at the same level.

$$p = \frac{F_1}{a} = \frac{F_2}{A} \text{ or } F_2 = \left(\frac{A}{a}\right) F_1$$

- ★ Consequently, the force on the larger piston is large. Thus, a small force F_1 acting on a small area a results in a larger force F_2 acting on a larger area A .



- Q.** A weighted piston confines a fluid of density ρ in a closed container, as shown in the figure. The combined weight of piston and weight is $W = 200 \text{ N}$, and the cross-sectional area of the piston is $A = 8 \text{ cm}^2$. Find the total pressure at point B if the fluid is mercury and $h = 25 \text{ cm}$ ($\rho_m = 13600 \text{ kg/m}^3$). What would an ordinary pressure gauge read at B?



- Sol.** Notice what Pascal's principle tells us about the pressure applied to the fluid by the piston and atmosphere. This added pressure is applied at all points within the fluid. Therefore, the total pressure at B is composed of three parts:

$$\text{Pressure of atmosphere} = 1.0 \times 10^5 \text{ Pa}$$

$$\text{Pressure due to piston and weight}$$

$$= \frac{W}{A} = \frac{200 \text{ N}}{8 \times 10^{-4} \text{ m}^2} = 2.5 \times 10^5 \text{ Pa}$$

$$\text{Pressure due to height } h \text{ of fluid} = \rho g h$$

$$= 0.33 \times 10^5 \text{ Pa}$$

In this case, the pressure of the fluid itself is relatively small. We have

$$\text{Total pressure at B} = 3.83 \times 10^5 \text{ Pa} = 383 \text{ kPa}$$

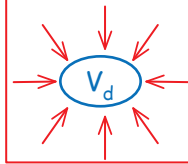
The gauge pressure does not include atmospheric pressure. Therefore,

$$\text{Gauge pressure at B} = 283 \text{ kPa}$$

BOUYANT FORCE

If a body is partially or wholly immersed in a fluid, it experiences an upward force due to the fluid surrounding it.

This phenomenon of force exerted by fluid on the body is called buoyancy and the force is called buoyant force.



Archimedes' Principle

A body immersed in a fluid experiences an upward buoyant force equivalent to the weight of the fluid displaced by it.

$$B = \rho_l V_d g$$

→ wt of the displaced liq by body

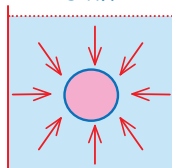
SKC

Body ko paani mein duboya toh paani ne pressure difference ki wajah se Force lagaya jiska sum kaddu (zero) hai. Inn vector ke sum ko Buoyant force kehte hain. Jiski value $\rho_l V_d g$ hai

Which is independent of P_0 , and distance of object from the free surface. Ab bata, yeh easy hai ya tough.



sink



If a body is fully inside water then it may have acc downward, upward or zero depending on value of density of liquid and body and volume.

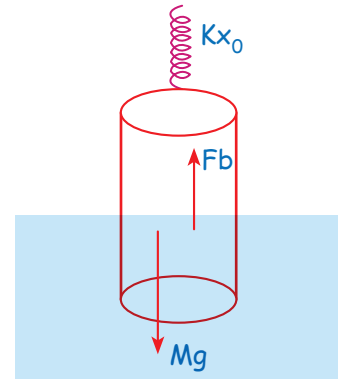
	If $B > mg$ acc upar gubbara ud gya (balloon in H_2O)
	If $mg > B$ acc niche (coin in water)

Q. A uniform cylinder of length L and mass M having cross-sectional area A is suspended, with its length vertical, from a fixed point by a massless spring, such that it is half submerged in a liquid of density σ at equilibrium position. The extension x_0 of the spring when it is in equilibrium is:

Sol. At equilibrium

$$Kx_0 = Mg - B$$

$$Kx_0 = Mg - \frac{\sigma AL}{2}$$



$$x_0 = \frac{\left(Mg - \frac{\sigma AL}{2}g\right)}{K}$$

$$= \left[1 - \frac{\sigma AL}{2M}\right]$$

Q. A solid sphere of radius R and density is attached to one end of a mass-less spring of force constant k . The other end of the spring is connected to another solid sphere of radius R and density 3ρ . The complete arrangement is placed in a liquid of density 2ρ and is allowed to reach equilibrium. The correct statement(s) is (are)

Sol. Weight of the balls i.e system = mg

$$= \left[\rho \times \frac{4}{3}\pi R^3 + 3\rho \times \frac{4}{3}\pi R^3\right]g$$

$$= \frac{16\rho\pi R^3 g}{3}$$

$$\text{Buoyant force} = V_{pg}$$

$$\rho_1 = 2\rho$$

$$= V = 2\rho g$$

$$\text{Buoyant force} = \text{weight of system}$$

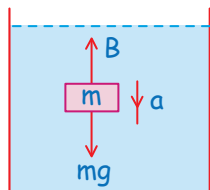
$$V \cdot 2\rho g = \frac{16\rho\pi R^3 g}{3}$$

$$V = \frac{8\pi R^3}{3}$$

$$\text{Submerged volume} = \frac{8\pi R^3}{3}$$

$$\text{Total volume of system} = \frac{4\pi R^3}{3} + \frac{4\pi R^3}{3}$$

- Q. Now if string is cut in above question so that, block is accelerating downward with accl a . Find acc of block, ($\rho_{\text{solid}} > \rho_{\text{liquid}}$)



Sol. $\rho_s > \rho_l \Rightarrow mg > B$

$$a = \frac{mg - B}{m} = \frac{\rho_s vg - \rho_l vg}{\rho_s v} = g - \frac{\rho_l}{\rho_s} g$$

Niche

$$a = g \left(1 - \frac{\rho_l}{\rho_s} \right)$$

Niche

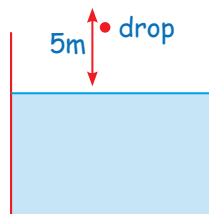
- Q. Repeat the above problem if $\rho_{\text{solid}} < \rho_{\text{liquid}}$

$$\rho_l > \rho_s \Rightarrow B > mg$$

$$a = \frac{B - mg}{m} = g \left(\frac{\rho_l}{\rho_s} - 1 \right)$$

Upar

- Q. A particle of density $\rho/2$ is dropped from a height of 5m from the liquid surface. Find maximum depth reached by particle. Assume energy loss during chhapak, density of liquid = ρ .



Sol. $V^2 = 0^2 + 2 \times 10 \times 5$
 $v = 10 \text{ m/s}$ (just before chhalang)
 When particle is in water

$$a = \frac{B - mg}{m} = g \left(\frac{\rho}{\rho/2} - 1 \right)$$

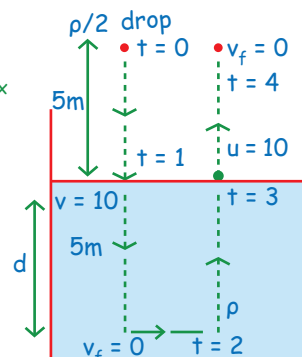
$$a = g \text{ (upar)}$$

$$0^2 = 10^2 - 2 \times 10 \times d_{\text{max}}$$

$$d_{\text{max}} = 5\text{m}$$

$$h = 5$$

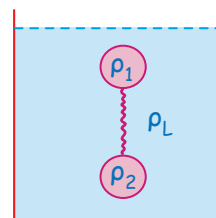
$$t = \sqrt{\frac{2h}{g}}$$



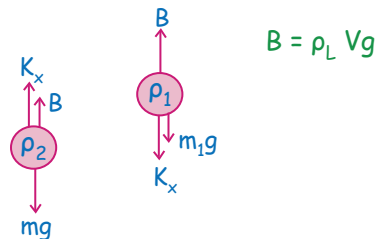
$$\text{Oscillation} = \text{SHM } \vec{F} = -kx$$

$$\text{Time period} = 1 + 1 + 1 + 1 = 4 \text{ sec}$$

- Q. Two bodies of same volume is attached through a spring and dipped into the water as shown. If are density of bodies and is density of the liquid then find the condition on which they will be in equilibrium. Also, draw the FBD or the bodies.



Sol. For Equilibrium, $\rho_1 < \rho_L < \rho_2$



- Q. A block of mass m, ρ, v having weight 50 N in air & 30 N in liquid of density ρ_L

- (1) Find buoyant force
 (2) Find ρ of liquid:

Sol. $B = 20 \text{ N}$

$$mg = 50 \Rightarrow \rho_L vg = 50$$

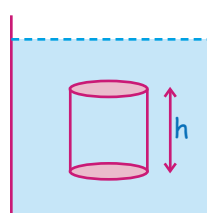
$$\rho_L Vg = 20$$

$$\rho_L \frac{50}{\rho_s} = 20$$

$$vg = 50/\rho_s$$

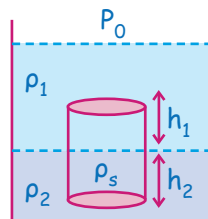
$$\rho_l = 2\rho_s/5$$

Q.



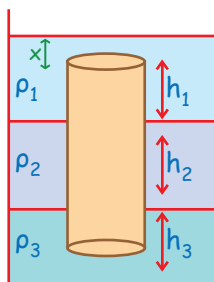
Sol. $B = V\rho_1 g$
 $B = \rho_2 Ahg$

- Q. A solid cylinder of density ρ_s as is immersed into two immiscible liquids as shown. Find the net force exerted by liquid on the cylinder.



Ans. $B_{\text{net}} = \rho_1 A h_1 g + \rho_2 A h_2 g$

- Q. Find the net force exerted by liquid on the cylinder.



agar ye cylinder equilibrium me toh
 $B_{\text{net}} = mg$



Ans. $B_{\text{net}} = \rho_1 A h_1 g + \rho_2 A h_2 g + \rho_3 A h_3 g$

FLOATING



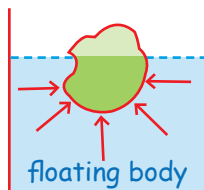
$$F_{\text{net}} = 0$$

$$mg = B$$

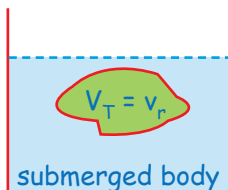
$$mg = \rho_L V_d g$$

$$V_d = \frac{m}{\rho_L}$$

independent of g_{eff} (in lift problem also)



floating body



submerged body

$$V_d = \text{pura ka pura Vol} = \text{vol}^n \text{ of body} = \frac{m}{\rho_s}$$

- Q. Find the force applied by fluid on slant surface solid cone having base radius r and Height h .

Sol.

$$B = \vec{F}_1 + \vec{F}_2$$

upar niche

$$B = F_2 - F_1$$

$$\rho_L V_d g = \rho g h \pi r^2 - F_1$$

$$\rho \frac{1}{3} \pi r^2 h g = \rho g h \pi r^2 - F_1$$

$$F_1 = \rho g h \pi r^2 \times 2/3$$

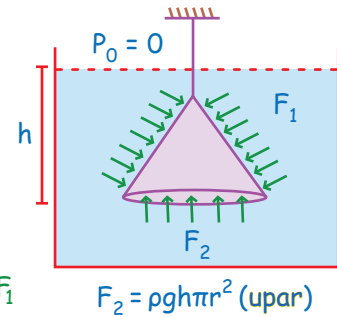
m - 2 SKC

assume cylinder

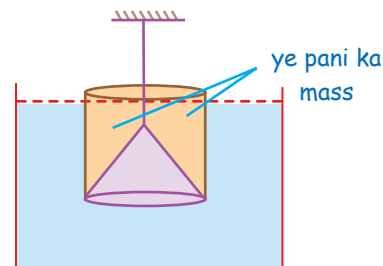
$$mg = (\pi r^2 h - 1/3 \pi r^2 h) \times \rho g$$

↘ cylinder ↘ cone

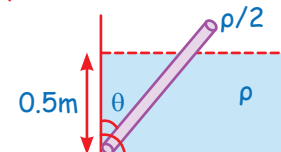
$$F_1 = 2/3 \pi r^2 h \times \rho g$$



$$F_2 = \rho g h \pi r^2 \text{ (upar)}$$



- Q. A rod of length 1m & density $\rho/2$ is in equilibrium as shown in the figure. Find the angle θ if density of liquid ρ .



- Sol. Let the cross-sectional area of rod be A

Rod is in equilibrium

$$B = \rho A x g$$

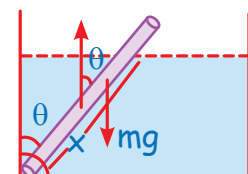
Point of application of buoyant force is com of displaced liquid.

Balancing to equal about hinge point

$$mg \sin \theta \times \frac{L}{2} = B \sin \theta \times \frac{x}{2}$$

$$\frac{\rho}{2} A L \sin \theta \times \frac{L}{2} = \rho A x g \sin \theta \times \frac{x}{2}$$

$$\frac{L^2}{2} = x^2 \text{ (L = 1m) given}$$



$$x = \frac{1}{\sqrt{2}}$$

$$\cos\theta = \frac{0.5}{x} = \frac{1}{2 \times \frac{1}{\sqrt{2}}} = \frac{1}{\sqrt{2}}$$

$$\theta = 45^\circ$$

Q. A cylinder of cross-sectional area A , length l , and density $\rho/2$ is floating in a liquid of density ρ as shown in the figure.

(1) Find q the length of cylinder (l') immersed in liquid.

Sol. $mg = B$

$$\rho_s A l g = \rho_l V_d g$$

$$\rho/2 \times A \times l \times g = \rho \times A \times l' \times g$$

$$l' = l/2$$

(2) If this system is placed in lift moving upward with acc ' a ' find l' :

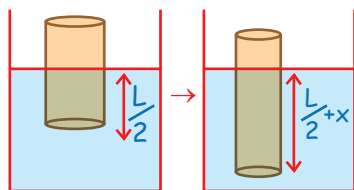
Sol. $mg_{\text{eff}} = \rho_l V_d g_{\text{eff}}$

$$m = \rho_l V_d$$

$$\rho/2 \times A l = \rho \times A \times l'$$

$$(l' = l/2 \rightarrow \text{same})$$

(3) Find the time period of oscillation if cylinder is displacement vertically downward by x



Sol. Initially $mg = B = \rho A \frac{l}{2} g$ (At equilibrium)

When displaced downward:

$$F_{\text{net}} = B - mg \text{ (upar)}$$

$$= A \rho \left(\frac{l}{2} + x \right) g - mg$$

$$\rho A l / 2 g + A \rho g x - mg$$

$$F_{\text{net}} = A \rho g x$$

$$\vec{F}_{\text{net}} = -A \rho g x \text{ (-ve sign is because force and displacement are in opposite direction)}$$

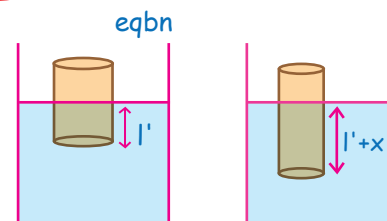
SHM:

$$T = 2\pi \sqrt{m/k}$$

$$T = 2\pi \sqrt{\frac{\rho_s A L}{A \rho_l g}}$$

$$T = 2\pi \sqrt{\frac{L}{g} \frac{\rho_s}{\rho_l}}$$

m - 2



$$mg = B \quad \vec{F}_{\text{net}} = B_{\text{extra}}$$

$$= \rho_l A x g$$

$$\vec{F}_{\text{net}} = \rho_l A g x$$

$$\vec{F} = -kx$$

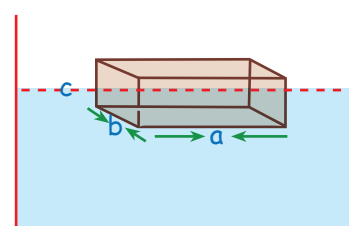
$$T = 2\pi \sqrt{\frac{m}{k}} = 2\pi \sqrt{\frac{\rho_s A L}{\rho_l A g}}$$

$$T = 2\pi \sqrt{\frac{L \rho_s}{g \rho_l}}$$

(4) cylinder ki jgah cube hota toh $T = ?$

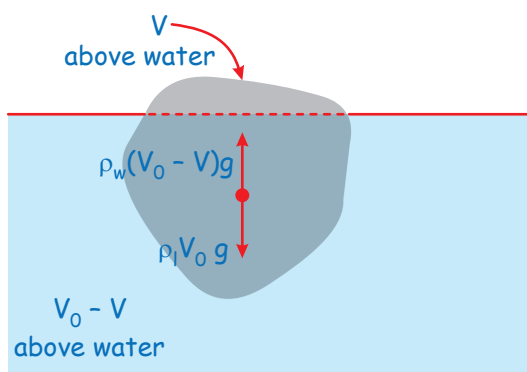
Sol. $T = 2\pi \sqrt{\frac{l}{g} \frac{\rho_s}{\rho_l}}$

(5) Cylinder ki jgah cuboid ho toh $T = ?$



Sol. $T = 2\pi \sqrt{\frac{C}{g} \frac{\rho_s}{\rho_l}}$

Q. An iceberg with a density of 920 kg m^{-3} floats on an ocean of density 1025 kg m^{-3} . What fraction of the iceberg is visible?



Sol. Let V be the volume of the iceberg above the water surface, then the volume under water will be $V_0 - V$. Under floating conditions, the weight ($\rho_i V_0 g$) of the iceberg is balanced by the buoyant force $\rho_w (V_0 - V)g$. Thus,

$$\rho_i V_0 g = \rho_w (V_0 - V)g$$

$$\text{or } \rho_w V = (\rho_w - \rho_i)V_0$$

$$\text{or } \frac{V}{V_0} = \left(\frac{\rho_w - \rho_i}{\rho_w} \right)$$

Since $\rho_w = 1025 \text{ kg m}^{-3}$ and $\rho_i = 920 \text{ kg m}^{-3}$, therefore,

$$\frac{V}{V_0} = \frac{1025 - 920}{1025} = 0.10$$

Hence 10 % of the total volume is visible.

Q. When a 2.5 kg crown is immersed in water, it has an apparent weight of 22 N. What is the density of the crown?

Sol. Let W = actual weight of the crown

W' = apparent weight of the crown

ρ = density of crown

ρ_o = density of water

The buoyant force is given by

$$F_b = W - W' \text{ or } \rho_o Vg = W - W'$$

$$\text{Since } W = \rho Vg, \text{ therefore, } V = \frac{W}{\rho g}$$

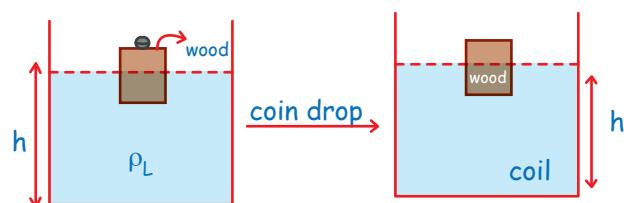
Eliminating V from the above two equations, we get

$$\rho = \frac{\rho_o W}{W - W'}$$

Here $W = 25 \text{ N}$; $W' = 22 \text{ N}$; $\rho_o = 10^3 \text{ kg m}^{-3}$

$$\therefore \rho = \frac{(10^3)(25)}{25 - 22} = 8.3 \times 10^3 \text{ kg m}^{-3}$$

Q. A coin is kept on top of a wooden block with is floating in a liquid as shown in the first figure given below. Now the coin is dropped into the liquid. Can you wherter height of liquid level will rise or fall or remains same?



$$\text{Sol. } (V_{\text{total}})_i = V_w + \frac{m_{\text{wood}} + m_{\text{coin}}}{\rho_L}$$

$$(V_T)_f = V_w + \frac{m_{\text{wood}}}{\rho_L} + \frac{m_{\text{coin}}}{\rho_{\text{coin}}}$$

$$\rho_{\text{coin}} > \rho_L$$

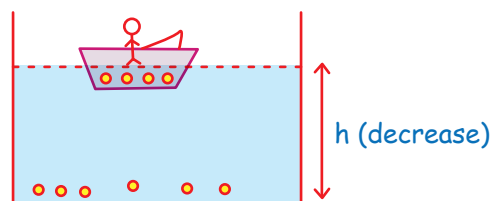
$$(V_T)_f < (V_T)_{\text{initial}}$$

$$\Rightarrow h' < h \quad (h \text{ decrease})$$

Agar coin ki jagah football rehta toh h (height) same hoga

★ Few examples of height decrease or increase:

(1) If man throw the stone into the water



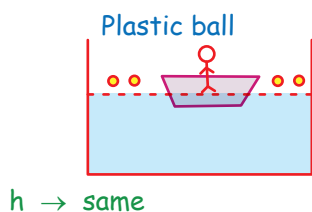
Sir, humne jo bachpan mein kauwa waali kahaani padhi thi, usme toh kauwa ne jab patthar daala tha toh paani ka level badh gaya tha, aur yahan kam ho raha hai.



Arey yaar, woh kauwa tumse zyada samajhdar hai, bahar se patthar laake paani mein daal raha hai... aur yahan pehle patthar naav mein indirectly tair rahe the aur phir doob gaye. Dono case alag hain.



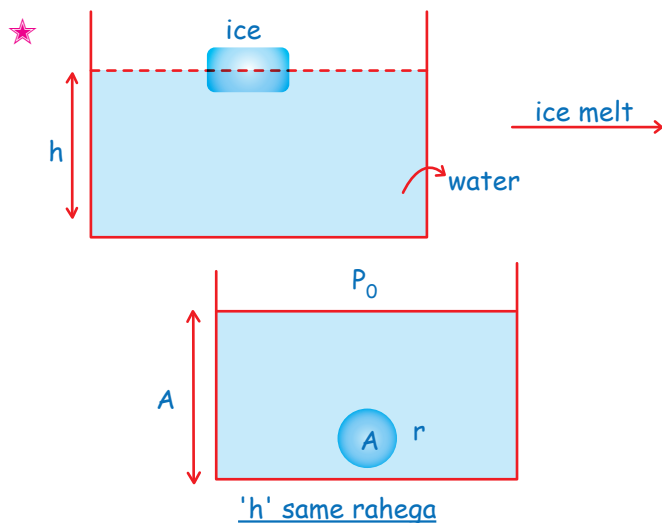
(2) If man throw the plastic ball into the water



SKC

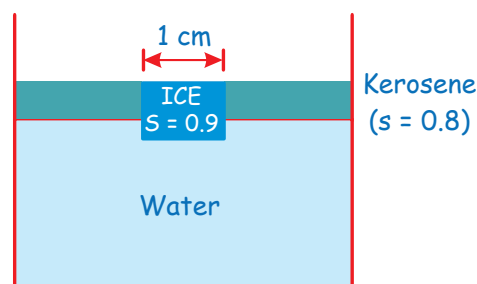
★ Floating ho gaya toh $h \rightarrow$ same

★ Sink ho gaya $h \rightarrow$ decrease



Q. An ice cube of side 1 cm is floating at the interface of kerosene and water in a beaker of base area 10 cm^2 . The level of kerosene is just covering the top surface of the ice cube.

- Find the depth of submergence in the kerosene and that in the water.
- Find the change in the total level of the liquid when the whole ice melts into water.



Sol. (a) Condition of floating

$$0.8 \rho_w g h_k + \rho_w g h_w = 0.9 \rho_w g h$$

$$\text{or } 0.8 h_k + h_w = 0.9 h \quad \dots (i)$$

where h_k and h_w be the submerged depth of the ice in the kerosene and water, respectively.

$$\text{Also, } h_k + h_w = h \quad \dots (ii)$$

Solving equations (i) and (ii), we get

$$h_k = 0.5 \text{ cm,}$$

$$h_w = 0.5 \text{ cm}$$

$$(b) 1 \text{ cm}^3 \xrightarrow{\text{melts}} 0.9 \text{ cm}^3$$

(Ice) (Water)

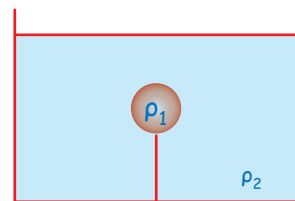
$$\text{Fall in the level of kerosene } \Delta h_k = \frac{0.5}{A}$$

$$\text{Rise in the level of water } \Delta h_w = \frac{0.9 - 0.5}{A} = \frac{0.4}{A}$$

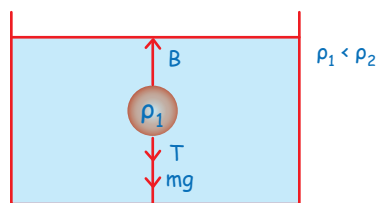
Net fall in the overall level

$$\Delta h = \frac{0.1}{A} = \frac{0.1}{10} = 0.01 \text{ cm} = 0.1 \text{ mm}$$

Q. A ball of density ρ_1 is attached to a string as shown in figure. If density of liquid is ρ_2 ($\rho_2 > \rho_1$) then find the tension in the string.



Sol. $B = T + mg$



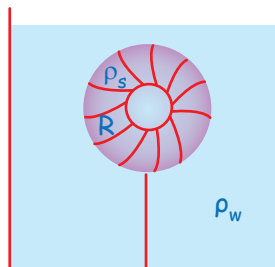
$$T = B - mg \Rightarrow T = \rho_2 v g - \rho_1 v g$$

Q. A hollow sphere is attached to a mass less string in a beaker as shown in figure.

density of sphere = ρ_L .

density of liquid = ρ_w , inner and outer radius are $R/2$ & R respectively.

(1) Tension in string = ?



Sol. $B = T + mg$

$$T = B - mg = \rho_w g \frac{4}{3} \pi R^3 - \rho_s$$

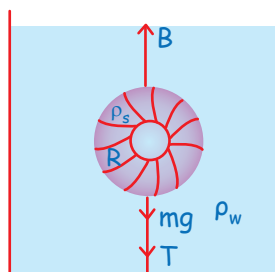
$$\left[\frac{4}{3} \pi R^3 - \frac{4}{3} \pi (R/2)^3 \right]$$

2. If cavity inside filled with liq of density ρ_1 find tension.

Sol.

$$\rho_1 < \rho_w$$

$$\rho_s < \rho_w$$

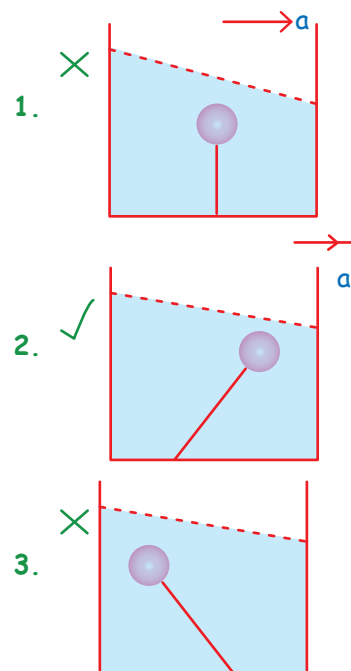
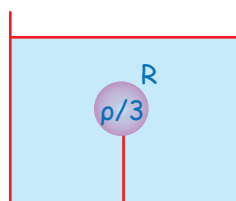


$$B = mg + T$$

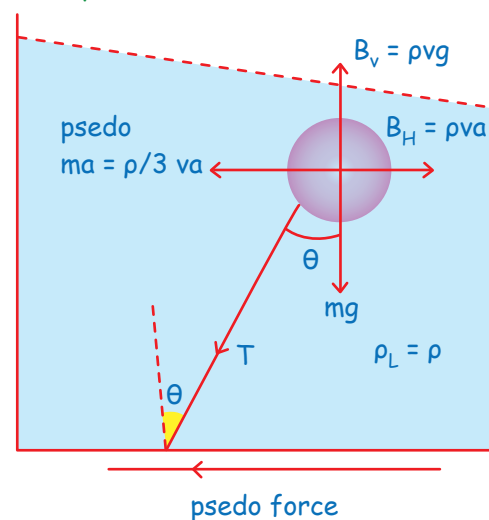
$$T = B - mg = \rho_w \frac{4}{3} \pi R^3 g - m_{\text{body}}$$

$$m_{\text{body}} = \rho_s \times \left[\frac{4}{3} \pi R^3 - \frac{4}{3} \pi (R/2)^3 \right] + \rho_1 \frac{4}{3} \pi (R/2)^3$$

Q. If beaker is given horizontal acceleration 'a' to the right side Diagram will be?



4. $\times$ Depends on value of 'a'.



$$T \cos \theta + mg = B_v$$

$$T \sin \theta + ma = B_H$$

SKC

Bhai, jaise pehle neeche jaane par pressure badhta tha, aur phir jab humne body rakhi toh uss par pressure difference ki wajah se buoyant force aaya, similarly yahan horizontal mein pressure ka difference hai. Left side pressure badh raha hai, so, iski wajah se right side mein horizontal buoyant force jaisa force aur aa jaayega. Vertical toh tha hi, ab horizontal mein B_H lgega

FLUID DYNAMICS

Ideal Fluid in Motion

★ **Steady flow (laminar flow):** Velocity of moving fluid at any fix point does not change with time. If every point of a steadily flowing liquid follows exactly the same path that has been followed by the particles preceding it, the flow is said to be streamlined.

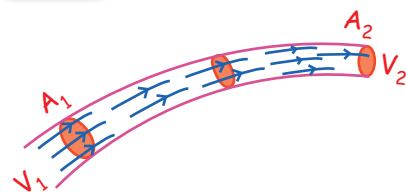
★ **Incompressible flow** $\rightarrow (\rho \rightarrow \text{uniform})$

★ Non Viscous flow

★ Irrotational flow

★ Each particle follow the same path as taken by previous particle passing through that point

EQUATION OF CONTINUITY



$$\frac{dm}{dt} = \rho A v$$

*** IMP

$$\left(\frac{dm}{dt}\right)_{in} = \left(\frac{dm}{dt}\right)_{out}$$

$(dm)_{in}$ = mass of water entering in 'dt' time = $\rho v_1 A_1 (dt)$

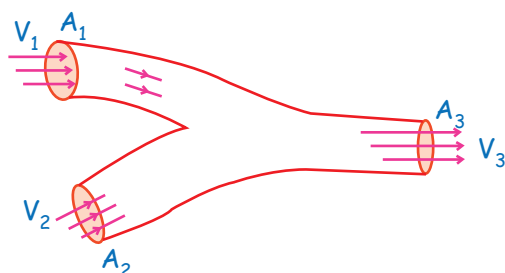
$$\left(\frac{dm}{dt}\right)_{in} = \rho A_1 v_1 \quad \left(\frac{dm}{dt}\right)_{out} = \rho A_2 v_2$$

$$\Rightarrow \rho A_1 v_1 = \rho A_2 v_2$$

$$A_1 v_1 = A_2 v_2$$

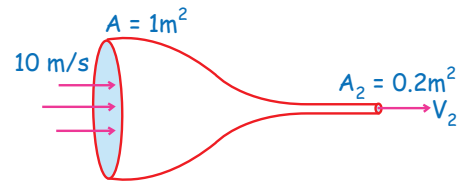
*** IMP

Q. Write the continuity equation for the given diagram.



$$A_1 v_1 + A_2 v_2 = A_3 v_3$$

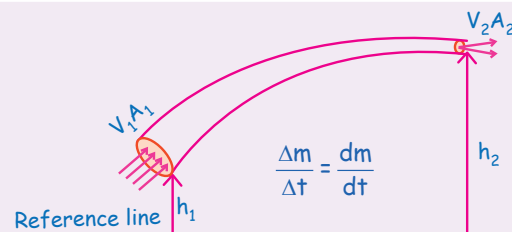
Q. Find the velocity V_2 for the given diagram.



$$A_1 v_1 = A_2 v_2$$

$$1 \times 10 = 0.2 \times V_2 = 50 \text{ m/s}$$

BERNOULLI THEOREM



$$P + \rho gh + \frac{1}{2} \rho v^2 = \text{const}$$

$$P_A + \rho gh_1 + \frac{1}{2} \rho v_1^2 = P_B + \rho gh_2 + \frac{1}{2} \rho v_2^2$$

Yeh yaad rakhna ki yahin jo ρgh hai woh purane wala pressure wala ρgh nahi hai yeh WPE mein potential energy mgh ka tau (uncle) hai, bole toh ussi ka rishtedaar hai. Jo oopar jaane par badhega. So, Bernoulli likhne se pehle hamesha tau ki line (reference line) bana lo.



Proof (Not important)



Ignore Ignore

Work Energy theorem work done by all the force = ΔKE

$$P_1 A_1 v_1 \Delta t - (\Delta m) g (h_2 - h_1) - P_2 A_2 v_2 \Delta t$$

$$= \frac{1}{2} \Delta m (v_2^2 - v_1^2)$$

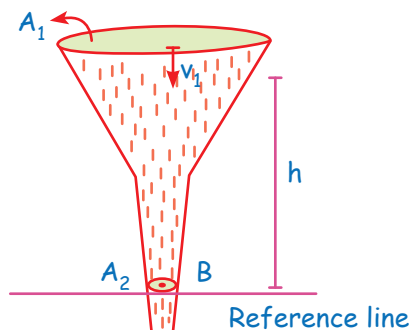
$$P_1 A_1 v_1 - \left(\frac{\Delta m}{\Delta t}\right) g (h_2 - h_1) - P_2 A_2 v_2 = \frac{1}{2} \left(\frac{\Delta m}{\Delta t}\right) (v_2^2 - v_1^2)$$

$$P_1 A_1 v_1 - \rho A_1 v_1 g (h_2 - h_1) - P_2 A_2 v_2$$

$$= \frac{1}{2} \rho (A_1 v_1) (v_2^2 - v_1^2)$$

$$P + \rho gh + \frac{1}{2} \rho v^2 = \text{const}$$

Q. Find the preservative point B.



$$(1) P_A + \rho gh + 1/2 \rho v_1^2 = P_B + 0 + 1/2 \rho v_2^2$$



Kaam ka Dabba

$$\star A_1 V_1 = A_2 V_2$$

$$\star \frac{dm}{dt} = \rho A v = \text{Rate of flow of mass}$$

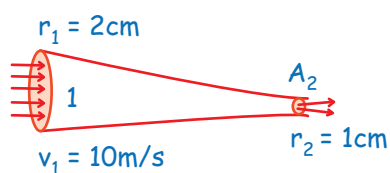
$$\star \frac{dvol^n}{dt} = A v = \text{volume flow rate}$$

$$\star P_1 + \rho gh_1 + 1/2 \rho v_1^2 = P_2 + \rho gh_2 + 1/2 \rho v_2^2$$

mg ka rishtedar
(Sabse pahle Reference line btao)

Next ques. is very important jo
JEE mein bhut bar pucha hai.

Q. Find pressure difference between points 1 and 2. Take $\rho = 1000 \text{ kg/m}^3$



$$(1) A_1 V_1 = A_2 V_2$$

$$\cancel{\pi r_1^2} V_1 = \cancel{\pi r_2^2} V_2$$

$$4 \times 10^{-4} \times 10 = 1 \times V_2 \times 10^{-4}$$

$$\Rightarrow V_2 = 40 \text{ m/s}$$

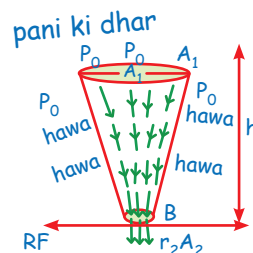
$$(2) P_1 + 0 + 1/2 \rho \times 10^2 = P_2 + 0 + 1/2 \rho (40)^2$$

$$P_1 - P_2 = 1/2 \rho \times 1500 = 1/2 \times 1000 \times 1500$$

$$P_1 - P_2 = 750 \text{ kPa}$$

Tap Water pani ki dhar wala Question

Q. Find the radius of pani ki dhar at a depth 'h' below point A & speed & volume rate flow



$$\text{Sol. } A_1 V_1 = A_2 V_2$$

$$P_A = P_0 ; P_B = P_0 \Rightarrow P_A = P_B$$

Apply Bernoulli theorem between A & B.

$$\cancel{P_B} + 0 + 1/2 \rho V_2^2 = \cancel{P_A} + \rho gh + 1/2 \rho V_1^2$$

$$V_2^2 = V_1^2 + 2gh$$



Dhyaan se dekho, paani ki dhaar jaise neeche jaane par patli hoti jaa rahi hai... Ab samajh aaya aisa kyun hua.... Neeche jaane par velocity badh rahi hai aur from eqn. of continuity $AV = \text{const} \Rightarrow \text{if } V \uparrow \text{ then } A \downarrow \Rightarrow r \downarrow$



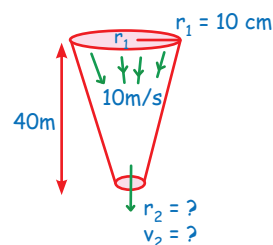
Bhai, yeh ques (question) JEE Mains mein 2-3 times poochha hai. Isme bas yeh do formula yaad kar lo jo ki bahut aasaan hain... Yeh bilkul motion under gravity mein third eqn (equation) of motion jaisa hai

$$V_2^2 = V_1^2 + 2gh$$

$$\frac{d vol^m}{dt} = A_1 V_1 = A_2 V_2$$



Q. Consider the given diagram. Liquid flows is water ($\rho = 1000 \text{ kg/m}^3$). Find V_2 & r_2 volume flow rate.



(1)

$$V_2^2 = 100 + 800$$

$$\Rightarrow V_2 = 30 \text{ m/s}$$

$$V_2^2 = V_1^2 + 2gh$$

(2)

$$A_1 V_1 = A_2 V_2$$

$$\cancel{\pi r_1^2} V_1 = \cancel{\pi r_2^2} V_2$$

$$(10)^2 \times 10 = r_2^2 \times 30$$

$$r_2 = \sqrt{\frac{1000}{30}} \text{ cm}$$

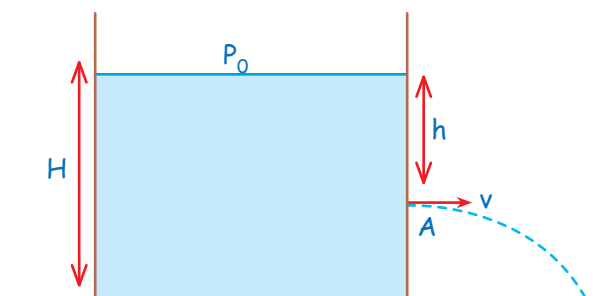
(3)

$$\frac{d\text{vol}^n}{dt} = A_1 V_1 = A_2 V_2$$

$$= \pi (0.1)^2 \times 10 = 0.1\pi \text{ m}^3/\text{s}$$

VELOCITY OF EFFLUX

bhai iski feel mai tumhe kya du ... iski feel to tum sare duniya ko subah sham na jane kitni bar dila sakte ho



If $A_{\text{hole}} \ll A_{\text{base}}$

$$A_1 V_1 = A_2 V_2 \therefore A_2 \ll A_1$$

$$\therefore V_2 \gg V_1$$

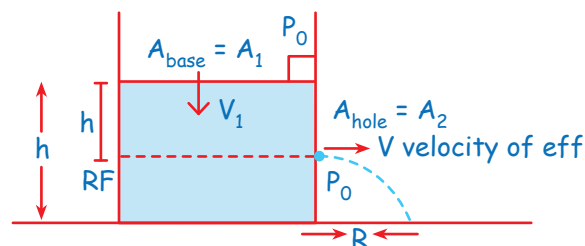
$$\cancel{P_0} + \cancel{pgh} + 1/2 \cancel{\rho V_1^2} = \cancel{P_0} + 0 + 1/2 \rho V_2^2$$

neglect

$$pgh = 1/2 (\rho V_2^2 - V_1^2)$$

(neglect V_1^2)

$$V_2 = \sqrt{2gh}$$



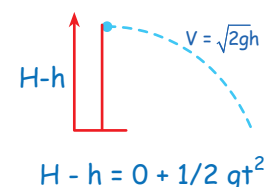
Range of Efflux

$$t = \sqrt{\frac{2(H-h)}{g}}$$

$$R = Vt$$

$$R = \sqrt{2gh} \sqrt{\frac{2(H-h)}{g}}$$

$$R = 2\sqrt{h(H-h)}$$



Q. For Range to be maximum, what should be h? And what will be the maximum Range?

Sol. For Range max $R = 2\sqrt{h(H-h)}$

$$R = 2\sqrt{h(H-h)}$$

max

$$h(H-h) \rightarrow \text{max}$$

$$(hH-h^2) \rightarrow \text{max}$$

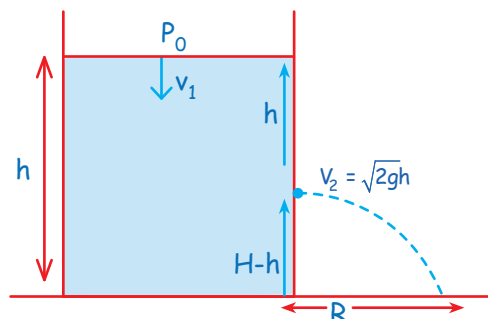
$$\frac{d}{dh}(hH-h^2) = 0$$

$$H - 2h = 0$$

$$h = \frac{H}{2}$$

$$R_{\text{max}} = 2\sqrt{\frac{H}{2}\left(H - \frac{H}{2}\right)} = H$$

Q. Consider the given diagram and answer the following questions.



(A) If system in lift

$$v = \sqrt{2g_{\text{eff}}h}$$

R → same

(B) If lift is free fall

$$g_{\text{eff}} = 0$$

$$v = \sqrt{2g_{\text{eff}}h} = 0$$

(C) Thrust force

$$= \frac{dm}{dt} v_{\text{rel}}$$

$$F = \rho A_2 V_2 \cdot V_2$$

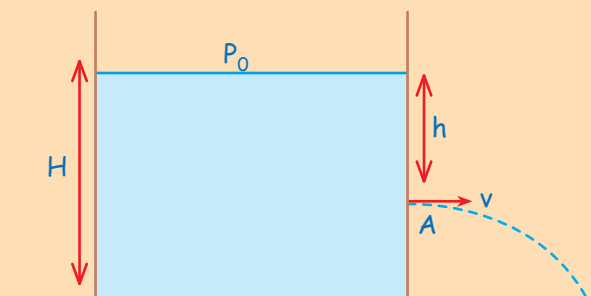
$$F = \rho A 2gh$$

(D) Find μ_{min} between ground & beaker due to thrust force. (beaker remains at rest)

$$\mu mg = F = \rho A 2gh$$



Kam ka Dabba



1 $V = \sqrt{2gh}$

2 $R = 2\sqrt{h(H-h)}$

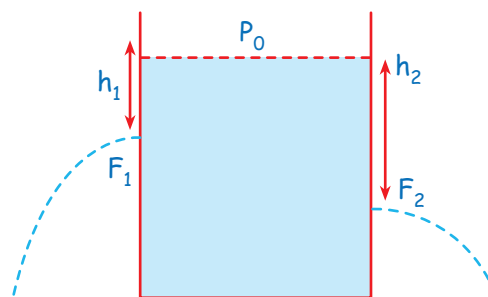
3 $R_{\text{max}} = H$ when $h = H/2$

4 Free fall = $v - 0$

5 Thrust force = $\rho A 2gh$

6 $\left[\frac{d(\text{vol})^n}{dt} \right]_{\text{out}} = A\sqrt{2gh}$

Q. If beaker experience net horizontal force of 20N Find $h_2 - h_1$, if area of hole is 10^{-3} m^2 each.
 $\rho \text{ water} = 1000 \text{ kg/m}^3$



Sol. $F_2 - F_1 = 20$

$$\rho A 2gh_2 - \rho A 2gh_1 = 20$$

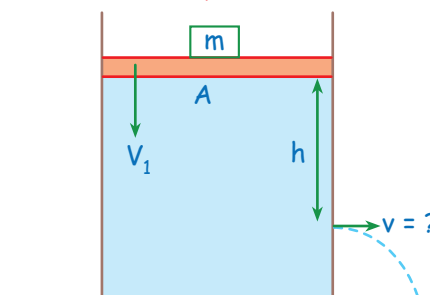
$$\rho A 2g (h_2 - h_1) = 20$$

$$10^{-3} \times 10^3 \times 2 \times 10 [h_2 - h_1] = 20$$

$$[h_2 - h_1] = 1$$

Q. Consider the given diagram and solve the following questions:

(a) Find velocity of efflux



$$A_{\text{hole}} \ll A_{\text{base}}$$

Solution

$$P_0 + \frac{mg}{A_{\text{base}}} + \rho gh + \frac{1}{2} \rho v_1^2 = P_0 + 0 + \frac{1}{2} \rho v_2^2$$

neglect

$$\frac{1}{2} \rho v_2^2 = \frac{mg}{A_{\text{base}}} + \rho gh$$

(b) If $\rho = 1000 \text{ kg/m}^3$, $m = 50 \text{ kg}$, $A_{\text{base}} = 1 \text{ m}^2$, $A_{\text{hole}} = 1 \text{ cm}^2$, $h = 2 \text{ m}$ then find V^2 .

$$\frac{1}{2} \times 1000 v_2^2 = \frac{500}{1} + 1000 \times 10 \times 2$$

$$500 v_2^2 = 20500$$

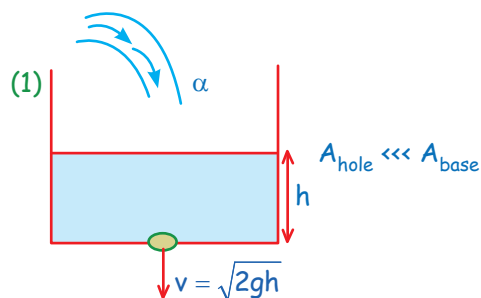
$$v_2 = \sqrt{41} \text{ m/s.}$$

Q. Suppose an empty bucket of base area A_{base} a hole of area A_{hole} in its base. If water is supplying to the bucket at a constant volume rate am^3/s then find:

(1) The level of water which becomes constant after long time.

(2) Graph 'h' vs 't'

Sol.

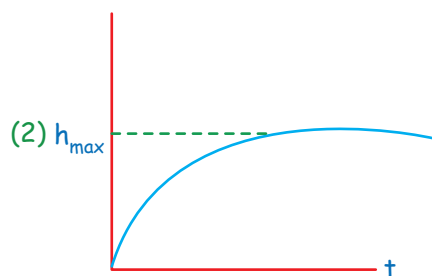


$$\frac{dh}{dt} = \frac{\alpha - A_{\text{hole}} \sqrt{2gh}}{A_{\text{base}}}$$

After very long time

$$h \rightarrow \text{const} \Rightarrow \frac{dh}{dt} = 0$$

$$\alpha = A_{\text{hole}} \sqrt{2gh} \Rightarrow h = \frac{\alpha^2}{2gA_{\text{hole}}^2}$$

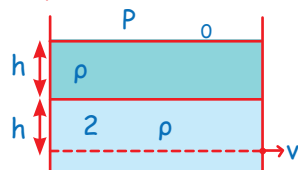


$$\alpha - A_{\text{hole}} \sqrt{2gh} = A_{\text{base}} \frac{dh}{dt}$$

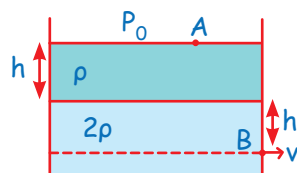
$$\int_0^h \frac{dh}{\alpha - A_{\text{hole}} \sqrt{2gh}} = \int_0^t \frac{dt}{A_{\text{base}}}$$

Isko solve karna 12th class mein sikhenge...

- Q. Two immiscible liquids of density ρ and 2ρ are kept in a beaker as shown in the figure. There is a hole at a height of $2h$ from the free surface. Find velocity of efflux.



Sol.



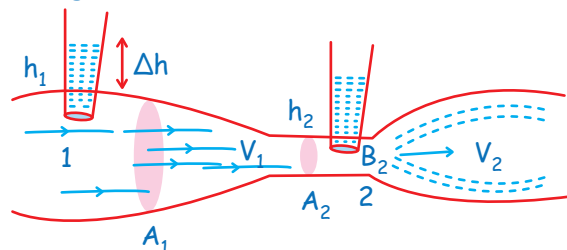
Applying Bernoulli's theorem between A and B:

$$P_0 + 2\rho gh + \rho gh + \frac{1}{2}\rho V_1^2 = P_0 + 0 + \frac{1}{2}2\rho V_2^2$$

$$V_2 = \sqrt{3gH}$$

VENTURI METER

It is used to find the rate of flow of fluid or velocity of flowing fluid.



$$P_1 = P_0 + \rho gh_1$$

$$P_2 = P_0 + \rho gh_2$$

$$P_1 - P_2 = \rho g(h_1 - h_2) = \rho g\Delta h \quad \dots(i)$$

Applying Bernoulli's theorem between 1 and 2:

$$P_1 + 0 + \frac{1}{2}\rho V_1^2 = P_2 + 0 + \frac{1}{2}\rho V_2^2$$

$$P_1 - P_2 = \frac{1}{2}\rho(V_2^2 - V_1^2) \quad \dots(ii)$$

from (i) and (ii)

$$\frac{1}{2}\rho(V_2^2 - V_1^2) = \rho g\Delta h$$

$$\therefore A_1 V_1 = A_2 V_2$$

$$\therefore \frac{1}{2}\rho V_2^2 - \left(\frac{A_2}{A_1} V_2\right)^2 = \rho g\Delta h$$

$$\frac{\rho V_2^2}{2} \left(1 - \frac{A_2^2}{A_1^2}\right) = \rho g\Delta h$$

$$V_2 = \sqrt{\frac{2g\Delta h}{1 - \frac{A_2^2}{A_1^2}}}$$

REYNOLDS NUMBER (R_N)

It is used to indicate to find laminar turbulent nature of flow

$$R = \frac{\rho v d}{\eta}$$

where ρ = density
 v = speed
 d = diameter pipe
 η = coeff of viscosity

Turbulent flow matlab utpatang flow

$R_N < 1000 \rightarrow$ laminar flow

$R_N > 2000 \rightarrow$ Turbulent flow

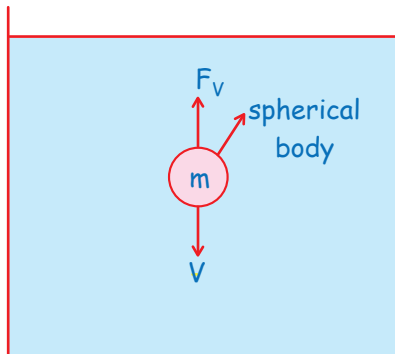
$1000 < R_N < 2000 \rightarrow$ Unsteady flow

STOKES LAW

When spherical body with velocity 'v' (relative to fluid) inside the fluid value of viscous force is given by.

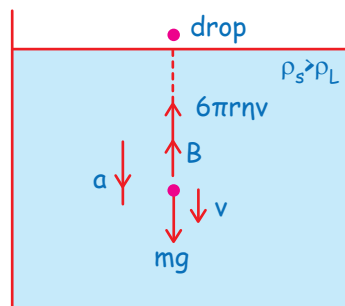
$$F_v = 6\pi r \eta v \quad (\text{Experimentally determined})$$

→ coeff of viscosity



$$\star F_{\text{net}} = mg - B - 6\pi r \eta v = ma$$

$$a = \frac{mg - B - 6\pi r \eta v}{m}$$



Terminal Velocity

$$F_{\text{net}} = 0$$

$$\Rightarrow mg = B + 6\pi r \eta v$$

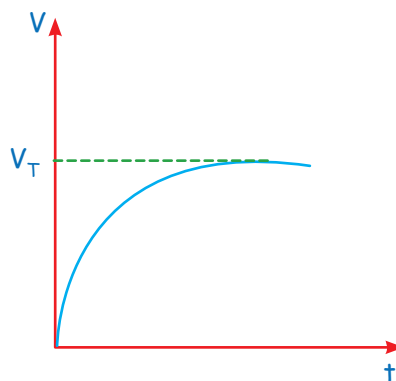
$$v = \frac{mg - B}{6\pi r \eta} = V_T$$

$$mg = \frac{4}{3} \pi r^3 \rho_s g$$

$$B = \frac{4}{3} \pi r^3 \rho_l g$$

$$V_T = \frac{mg - B}{6\pi r \eta}$$

$$V_T = \frac{4/3 \pi r^3 \rho_s g - 4/3 \pi r^3 \rho_l g}{6\pi r \eta} = \frac{2}{9} \frac{r^2 g}{\eta} (\rho_s - \rho_l)$$



$$V_T = \frac{2}{9} \frac{r^2 g}{\eta} (\rho_s - \rho_l)$$

This is very very important formula, jiss par bahut saare question Mains aur Advance mein pooche gaye hain. Mains mein toh har saal question aa hi jaata hai



Q. A stone of density ($\rho_s < \rho_l$) is release from surface of water at $t = 0$. Find its velocity as function of time, $v = f(t)$.

$$\text{Sol. } a = \frac{mg - B - 6\pi r \eta v}{m}$$

$$a = \left(\frac{mg - B}{m} \right) - \left(\frac{6\pi r \eta}{m} \right) v$$

$$a = k_1 - k_2 v$$

$$\int_0^v \frac{dv}{k_1 - k_2 v} = \int_0^t dt$$

$$\frac{-1}{k_2} \ln(k_1 - k_2 v) \Big|_0^v = t$$

$$\ln \left(\frac{k_1 - k_2 v}{k_1} \right) = -k_2 t$$

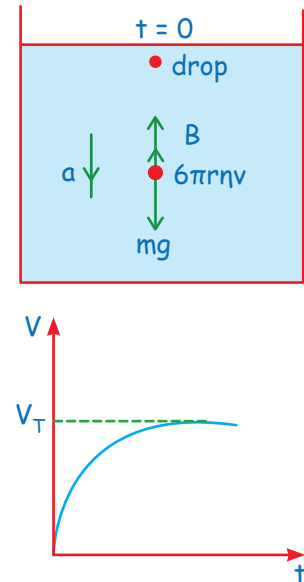
$$\left(\frac{k_1 - k_2 v}{k_1} \right) = e^{-k_2 t}$$

diff

$$\frac{k_1}{k_2} (1 - e^{-k_2 t}) = v$$

$$a = k_1 e^{-k_2 t}$$

$$t \rightarrow \infty \quad v = \frac{k_1}{k_2} (1 - 0) = \frac{mg - B}{6\pi r \eta} = V_T$$



VISCOSITY

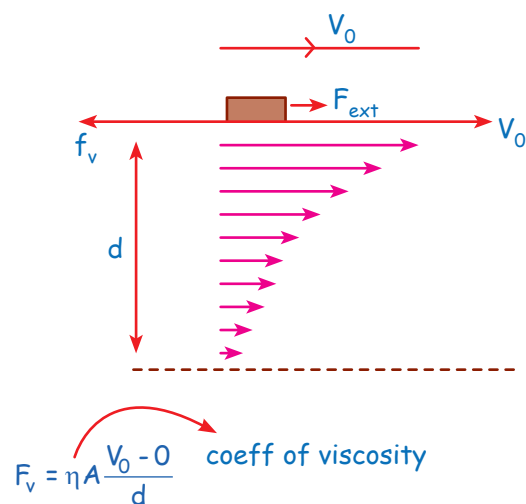
★ When a layer of a fluid slips or tends to slip on another layer in contact, the two layers exert tangential forces on each other. The directions are such that the relative motion between the layers is opposed, this property of a fluid to oppose relative motion between its layers is called viscosity. The forces between the layers opposing relative motion between them are known as the forces of viscosity.

- ★ Velocities of different layers are different.
- ★ The force of viscosity between two layers of a fluid is proportional to the velocity gradient in the direction perpendicular to the layers. Also the force is proportional to the area of the layer.

$$F_v \propto A \text{ (contact area)}$$

$$F_v \propto \frac{dv}{dx} \text{ (velocity gradient)}$$

$$\frac{dv}{dx} = \frac{V-0}{d}$$



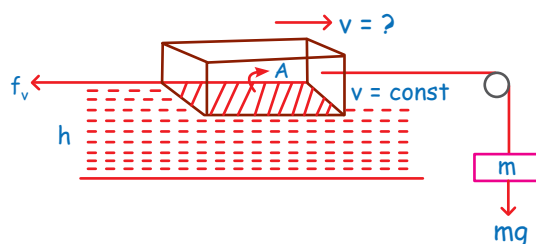
Bhai, yeh friction jaisa force hai, bas iss baar yeh area of contact par bhi depend kar raha hai. Iska formula galti se bhi mat bhoolna

$$f_v = \eta A \frac{dv}{dx}$$

$$\frac{f_v}{A} = \eta \frac{dv}{dx} = \text{shear stress}$$



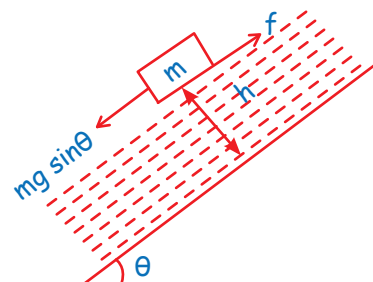
Q. Find the velocity of block.



$$\text{Sol. } mg = f_v = \eta A \frac{(v-0)}{h}$$

$$v = \frac{mgh}{\eta A}$$

Q. If block is moving down with constant velocity in following figure. Find the velocity of block.



$$\text{Sol. } mg \sin \theta = f_v$$

$$mg \sin \theta = \eta A \left(\frac{v-0}{h} \right)$$

Q. When two drop each of radius r with terminal velocity V_0 combined to form another spherical drop. Find new terminal velocity.

$$\text{Sol. } (Vol^n)_i = (Vol^n)_f$$

$$\frac{4}{3} \pi r^3 \times 2 = \frac{4}{3} \pi r_f^3 \Rightarrow r_f = 2^{1/3} r$$

$$V_t = \frac{2gr^2(\rho_s - \rho_l)}{9\eta} \quad \frac{V_0}{V_f} = \frac{r_1^2}{r_f^2}$$

$$V_t \propto r^2 \quad \frac{V_0}{V_f} = \frac{r^2}{2^{2/3} r^2}$$

$$V_f = 2^{2/3} V_0 = 4^{1/3} V_0$$

Q. n identical drop with terminal velocity V_0 combined to form a bigger size drop. Find the new terminal velocity.

$$\text{Sol. } \frac{4}{3} \pi r^3 \times n = \frac{4}{3} \pi r_f^3$$

$$r_f = n^{1/3} r$$

$$V_f = n^{2/3} V_0$$

SURFACE TENSION

Yeh liquid ke free surface ki property hai, jiski wajah se liquid ka free surface ek elastic stretch membrane ke jaisa behave karta hai... Samjha... Abe, yeh ek baar mein thode hi samajh aata hai... Neeche kuch images attach kar raha hoon, usse tumhe kuch-kuch idea lagega



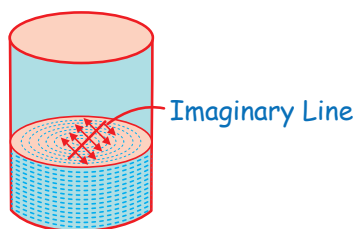


Kya tum tooti hui chai chhanni (strainer) ke beech soap film dekh paa rahe ho.... Issi mein phoonk maarne se bubble ban jaate hain



- ★ Surface tension is a property of liquid at rest by virtue of which a liquid surface gets contracted to a minimum area and behaves like a stretched membrane.
- ★ Surface tension of a liquid is measured by force per unit length on either side of any imaginary line drawn tangentially over the liquid surface, force being normal to the imaginary line as shown in figure i.e. surface tension.

$$(T) = \frac{\text{Total force on either side of the imaginary line (F)}}{\text{Length of the line (l)}}$$



Surface tension' (s)

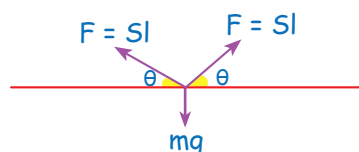
$$'S' = \frac{F}{l}$$

$$F = S \cdot \ell$$

- ★ Independent on location of imaginary line
- ★ It is property of free surface area

- ★ It is having tensile nature
- ★ For a given volume sphere has minimum surface area, that why all planets are approx spherical.
- ★ The surface tension of a liquid decreases with the rise in temperature and vice versa.

Q. Find the maximum mass of needle of length ℓ which can be placed on surface liquid without sink.

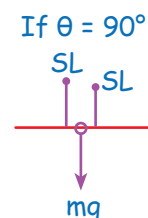


Sol. Free surface area $\uparrow$
 $2 S \ell \sin \theta = mg$

$$m = \frac{2 S \ell \sin \theta}{g}$$

If $\theta = 90^\circ$,

$$m = \frac{2 S \ell}{g}$$



Q. Find max mass of disc (m, R) which can be placed on liq free surface without sink

Sol. $S \cdot 2\pi R = mg$
 $\downarrow$ surface tension

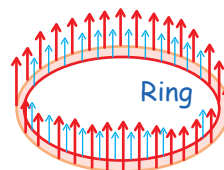
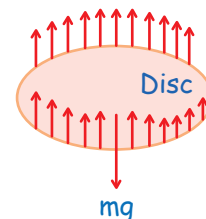
For Ring

$$S 2\pi R_{\text{बाहर}} + S 2\pi R_{\text{अंदर}} = mg$$

$$S 2\pi R_2 + S 2\pi R_1 = mg$$

$$R_1 = R_2 = R \quad (\because \text{thickness of ring is v small})$$

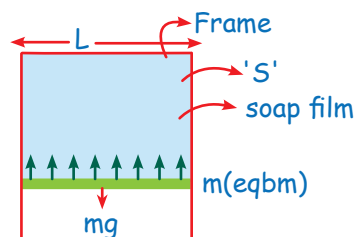
$$S 2\pi R \times 2 = mg$$



SKC

Surface tension ke dwaara Force aisa lagta hai ki liquid apna surface area min (minimum) karna chahe.

Q. (wooden rod m, L) is in eqbm find value of mass

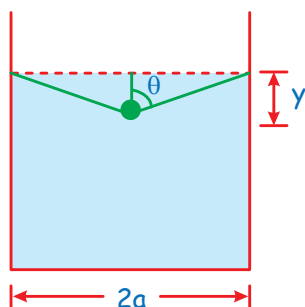


Sol. Soap film (2 layer)

$$2SL = mg$$

$$m = \frac{2SL}{g}$$

Q. A container of width $2a$ is filled with a liquid. A thin wire of weight per unit length λ is gently placed over the liquid surface in the middle of the surface as shown in the figure. As a result the liquid surface is depressed by a distance y ($y \ll a$). Determine the surface tension of the liquid.



Sol. By force balance on the wire,

$$2T \cos \theta \ell = \lambda \ell g$$

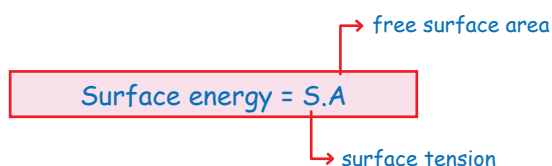
$$\frac{2Ty}{\sqrt{a^2 + y^2}} \ell = \lambda \ell g$$

Taking approximation

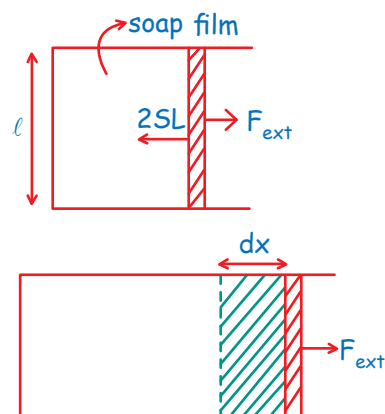
$$T = \frac{\lambda a g}{2y}$$

SURFACE ENERGY

★ The potential energy of surface molecules per unit area of the surface is called surface energy.



★ Suppose in following soap film we are applying external force F_{ext} such that wooden rod move by distance dx



★ Whatever work is being done by F_{ext} that goes to increase its surface energy.

$$dw_{ext} = F_{ext} \cdot dx = 2SL dx$$

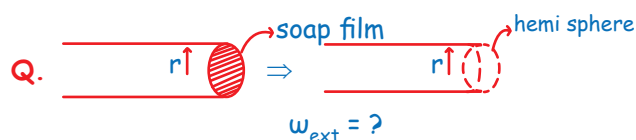
$$dw_{ext} = S \cdot 2(L dx) = S dA$$

$$\text{Hence } (WD)_{ext} = \Delta U$$

where $U = S.A = \text{surface energy}$

SKC

Ye formula agar tu bhul gaya matlab mujhe bhul gaya... Samajh gaya na.... S.A bole toh Saleem Ahmed bhi toh hota hai



$$(WD)_{ext} = \Delta(S.E) = (S.E)_f - (S.E)_i$$

$$= [S \times 2\pi R^2] \times 2 - [S \times \pi R^2] \times 2$$

Q. The surface tension of a soap solution is $3.5 \times 10^{-2} \text{ Nm}^{-1}$. The amount of WD require to increase the radius of soap bubble from 10 cm to 20 cm is?

$$\begin{aligned} \text{Sol. } (WD)_{ext} &= \Delta(S.E) = (S.E)_f - (S.E)_i \\ &= S \cdot 4\pi r_f^2 \times 2 - S \cdot 4\pi r_i^2 \times 2 \\ &= S \cdot 4\pi \times 2 [r_f^2 - r_i^2] \\ &= 3.5 \times 10^{-2} \times 4 \times 22/7 \times 2 [400 - 100] \times 10^{-4} \\ &= 88 \times 10^{-6} \times 300 \\ &= 264 \times 10^{-4} \text{ J} \end{aligned}$$

IMP

- Q. Find the work in transformation of a soap bubble of radius R_1 into a soap bubble of radius R_2 .

soap bubble $\xrightarrow{\quad}$ soap bubble
 R_1 R_2

Sol. $W_{ext} = (5.4\pi R_2^2 - 5.4\pi R_1^2) \gamma$

- Q. A liquid drop (spherical of Radius R is split into n small spherical drops.

(1) Find increase in surface Energy?



$$(\text{vol})_i = (\text{vol})_f$$

$$(4/3)\pi R^3 = n(4/3)\pi r^3$$

$$R = n^{1/3} r$$

$$r = n^{-1/3} R$$

$$(SE)_i = 5.4\pi R^2$$

$$(SE)_f = n4\pi r^2 \times \gamma$$

$$\Delta(SE) = (SE)_f - (SE)_i$$

$$\Delta(SE) = 5.4\pi n r^2 \times \gamma = 5.4\pi R^2$$

$$= 5.4\pi n^{-2/3} R^2 \cdot n - 5.4\pi R^2$$

$$= 5.4\pi R^2 n^{1/3} - 5.4\pi R^2$$

$$= 5.4\pi R^2 (n^{1/3} - 1)$$

2. Find decrease in temp in (1) part

$$\Delta(SE) = mS\Delta T$$

Specific heat

Decrease in internal energy = Increase in Surface Energy

3. Find change in surface energy of a bubble (not soap bubble) burst into 27 new smaller drops.



Sol. $4/3\pi R^3 = 27 \times 4/3\pi r^3$
 $R = 3r \Rightarrow r = R/3$

$$\Delta SE = 27 \times 5.4\pi r^2 - 5.4\pi R^2$$

$$= 27 \times 5.4\pi \times \frac{R^2}{9} - 5.4\pi R^2$$

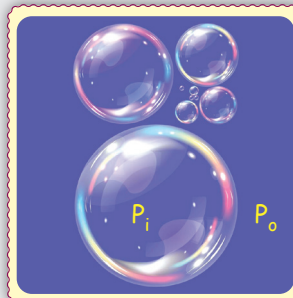
$$5.4\pi R^2 [3 - 1] = 10.8\pi R^2$$

This is very important question asked many times in JEE



IDEA OF EXCESS PRESSURE

Soap Bubble

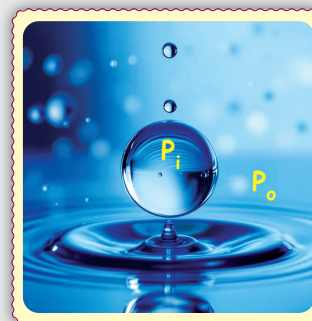


$$P_i - P_o = 4\gamma/R$$

$$(P_{\text{inside}} > P_{\text{outside}})$$

Matlab soap bubble ke andar agar jaoe to pressure $4\gamma/R$ badh jayega.

Liquid Drop (Boond)



$$P_i - P_o = 2\gamma/R$$

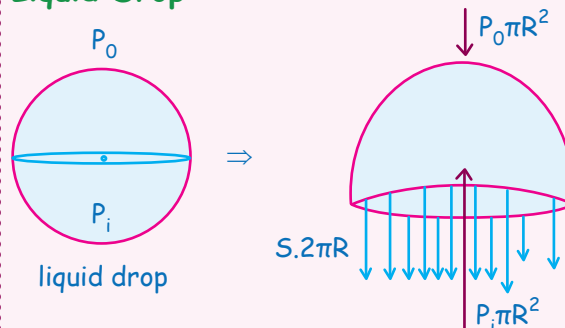
$$(P_{\text{inside}} > P_{\text{outside}})$$

Matlab soap bubble ke andar agar jaoe toh pressure $4\gamma/R$ badh jayega

★ Real life mein gravity ki effect ki wajah se liquid drop thoda elongated or oval shape ka ho jaata hai

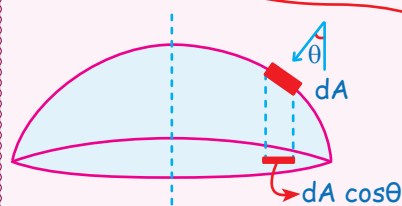
PROOF

Liquid Drop



$$P_i \pi R^2 = P_0 \pi R^2 + S \cdot 2\pi R$$

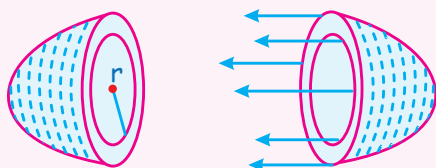
$$P_i - P_0 = \frac{2S}{R}$$



$$(F_{\text{net}}) \text{ by } P_0 = \int P_0 dA \cos \theta = P_0 \int dA \cos \theta = P_0 \pi R^2$$

Soap Bubble

Consider a soap bubble of radius r . Let p be the pressure inside the bubble and p_a outside. Imagine the bubble broken into two halves and consider one half of it as shown in figure. Since there are two surface, inner and outer, so the force due to surface tension is



$$F = \text{Surface tension} \times \text{Length} = S \times 2 (\text{circumference of the bubble}) = S \times 2 (2\pi r) \quad \dots (i)$$

$$\Rightarrow F = (p - p_a) \pi r^2. \quad \dots (ii)$$

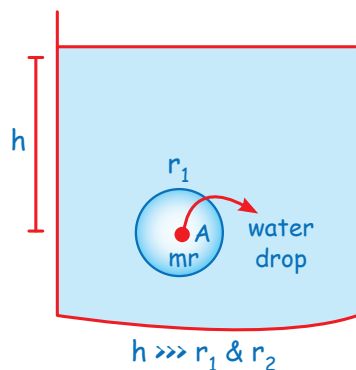
Maintain the equilibrium,

$$(p - p_a) \pi r^2 = S \times 2 (2\pi r)$$

$$\text{or } (p - p_a) = \frac{4S}{r}$$

$$\Rightarrow P_{\text{excess}} = \frac{4S}{r}$$

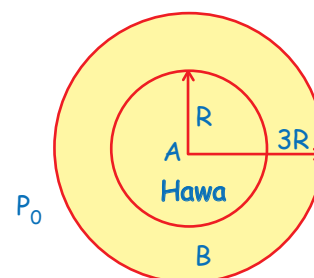
Q. Find P inside liquid drop.



$$h \gg r_1 \text{ \& } r_2$$

$$\text{Sol. } P_0 + \rho gh + \frac{2S}{r_1} + \frac{2S}{r_2} = P_A$$

Q. Find the ratio of gauge pressure at A and B.

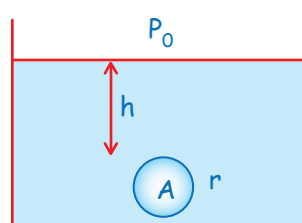


$$\text{Sol. } P_B = P_0 + \frac{2S}{R_{\text{bahar}}} = P_0 + \frac{2S}{3R}$$

$$P_A = P_0 + \frac{2S}{R_{\text{bahar}}} + \frac{2S}{R_{\text{andar}}} = P_0 + \frac{2S}{3R} + \frac{2S}{R}$$

$$\frac{(\text{gauge}) \text{ Press A}}{(\text{gauge}) \text{ Press B}} = \frac{\frac{2S}{3R} + \frac{2S}{R}}{\frac{2S}{3R}} = 4:1$$

Q. An air bubble of radius r is present at depth of h from free surface as shown in the figure. Find the pressure at the centre of the bubble if its radius is r .

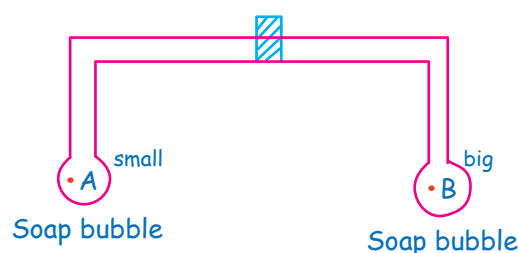


#SKC
gole ke
andar jao
toh Pressure
badhao



$$\text{Sol. } P_A = P_0 + \rho gh + \frac{2S}{r}$$

Q. Just after opening the valve air will flow from which side to where?



air flow
 $A \rightarrow B$

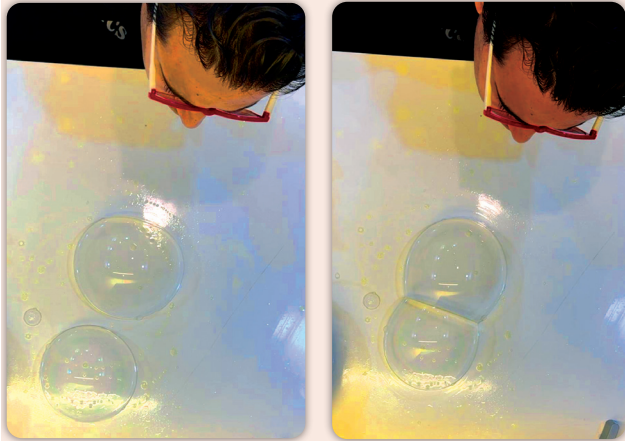
$$P_A = P_0 + \frac{4S}{R_A}$$

$$R_B > R_A$$

$$P_B < P_A$$

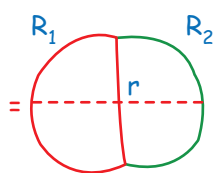
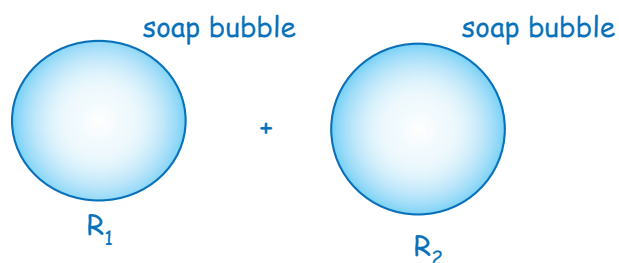
$$P_B = P_0 + \frac{4S}{R_B}$$

When two soap bubbles of radii r_1 and r_2 are attached as shown. Then the radius of curvature of the common film will be $r = \frac{R_1 R_2}{R_1 - R_2}$



I hope tumhe in dono image se feel aa gai hogi...
Abe sun iska derivation utna important nahi hai
jitna formula aur pichle 5 saalo mein ye ques. 4-5
baar Mains poochha hai

Proof

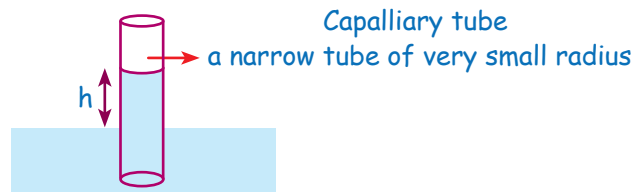
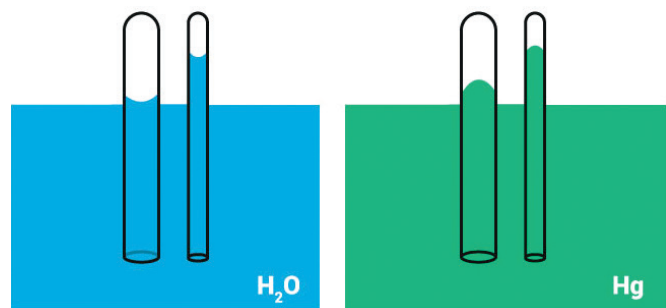


$$P_0 + \frac{4S}{R_1} + 0 + \frac{4S}{r} + 0 - \frac{4S}{R_2} = P_0$$

$$\frac{1}{r} = \frac{1}{R_2} - \frac{1}{R_1} = \frac{R_2 - R_1}{R_1 R_2}$$

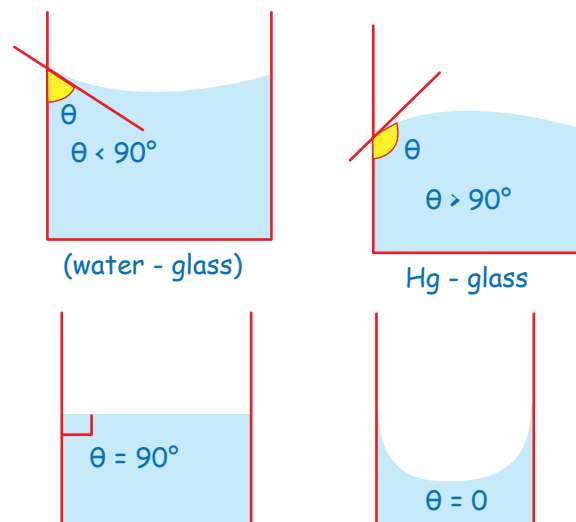
$$r = \frac{R_1 R_2}{R_1 - R_2}$$

Capillary Tube/Capillary Action



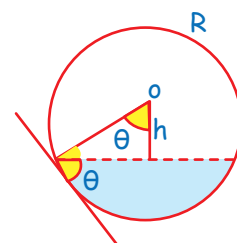
ANGLE OF CONTACT

The angle which the tangent to the liq surface at the point of contact makes with solid surface inside the liquid is called angle of contact.



liquid apna area net force ke $\perp$ ar rakhna chahta hai

Q. Find angle of contact.



Sol. $\cos\theta = \frac{h}{R}$

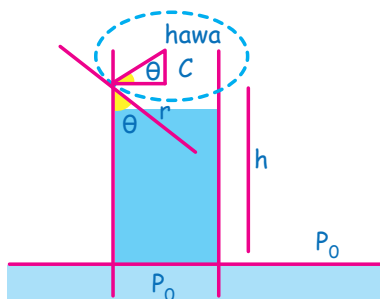
$\theta = \cos^{-1}(h/R)$

Height in the Capillary Tube

$h = \frac{2S\cos\theta}{r\rho g}$
 angle of contact (given)
 $r =$ radius of capillary tube
 $R =$ Radius of meniscus
 $h = \frac{2S}{R\rho g}$



Proof



$R \rightarrow$ Radius of meniscus

$$P_c = P_0 - \rho gh + \frac{2S}{R} = P_0$$

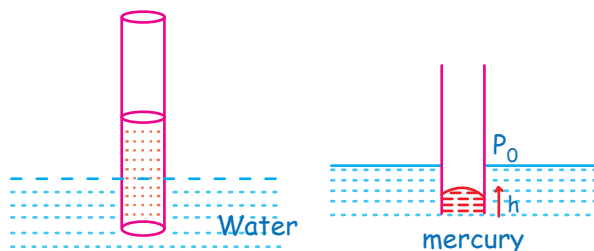
$$h = \frac{2S}{R\rho g}$$

$$R \cos \theta = r$$

$$R = r/\cos\theta$$

$$h = \frac{2S\cos\theta}{r\rho g}$$

Q. Consider the two systems of capillary tube one in liquid water and the other one in a mercury. Write down the expressions for rise/fall in each system.



Sol. For water: $h = \frac{2S\cos\theta}{r\rho g}$

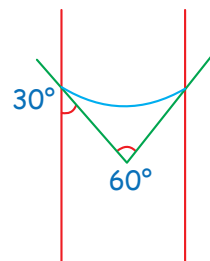
For mercury

$$P_0 + \rho gh - \frac{2S}{R} = P_0$$

$$h = \frac{2S}{R\rho g} \text{ or } h = \frac{2S\cos\theta}{r\rho g}$$

Q. A capillary tube made of glass of radius 0.15 mm is dipped vertically in a beaker filled with methylene iodide (surface tension = 0.05 Nm^{-1} , density = 667 kg m^{-3}) which rises to height h in the tube. It is observed that the two tangents drawn from liquid-glass interfaces (from opposite sides of the capillary) make an angle of 60° with one another. Then h is close to ($g = 10 \text{ ms}^{-2}$)

Sol. Angle of contact = 30°

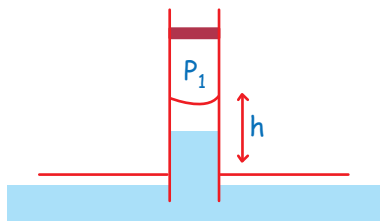


$$h = \frac{2T\cos\theta}{r\rho g} = \frac{2 \times 0.05 \times \left(\frac{\sqrt{3}}{2}\right)}{0.15 \times 10^{-3} \times 667 \times 10} = 0.087 \text{ m}$$

★ If tube is of insufficient length therefore the water will rise up to the upper end of the tube.

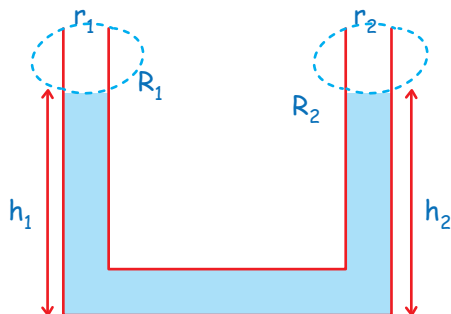
★ अगर main poore system ko free fall wali lift ke andar rakh doon toh $g_{\text{eff}} = 0$. Jitne length ki tube hai utna liquid rise karega

Q. Find the pressure P_1 for given diagram



Sol. $P_0 - \rho gh + \frac{2s}{R} = P_1$

Q. Find $h_1 - h_2$?



Sol. $P_0 - \frac{2s}{R_1} + \rho gh_1 + 0 - \rho gh_2 + \frac{2s}{R_2} = P_0$

$R_1, R_2 \rightarrow$ Radius of meniscus

$$P_0 - \frac{2s \cos \theta}{r_1} + \rho gh_1 + 0 - \rho gh_2 + \frac{2s \cos \theta}{r_2} = P_0$$

$h_1 - h_2 = \checkmark$

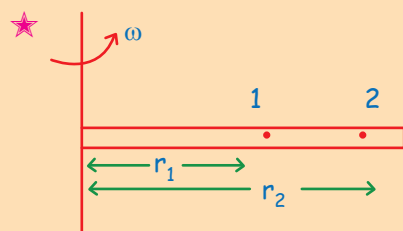


Kaam ka Dabba

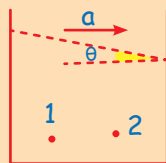
Bhai is chapter ke sabse zyada use hone wale saare formula main yahaan ek saath likh de raha hoon.... rough copy par likh-likh kar achhe se practice kar lena

★ $P_1 - P_2 = \rho a \ell$

$\tan \theta = a/g$



$$P_2 - P_1 = \frac{1}{2} \rho \omega^2 (r_2^2 - r_1^2)$$



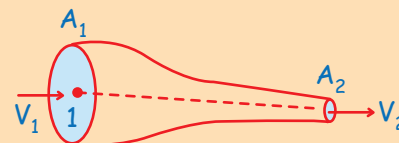
★ $B = \rho V_d g$

★ $mg = B = \rho_L V_d g$ (floating condition)

★ $A_1 V_1 = A_2 V_2$

$$\frac{dVol^n}{dt} = AV$$

$$\frac{dm}{dt} = \rho AV$$

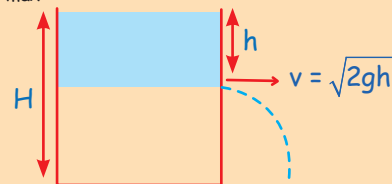


$$P + \rho gh + \frac{1}{2} \rho V^2 = \text{const}$$

$$P_1 + 0 + \frac{1}{2} \rho V_1^2 = P_2 + 0 + \frac{1}{2} \rho V_2^2$$

$$P_1 - P_2 = \frac{1}{2} \rho (V_2^2 - V_1^2)$$

★ $R_{\max} = H$ when $h = H/2$



$$R = 2\sqrt{h(H-h)}$$

$$F_{\text{thrust}} = \rho A_{\text{hole}} 2gh$$

★ Pani ki dhar tap water $V_2^2 = V_1^2 + 2gh$

★ Venturimeter

$$V_2 = \sqrt{\frac{2g\Delta h}{1 - \frac{A_2^2}{A_1^2}}}$$

★ Reynolds No

$$R_n = \frac{\rho V d}{\eta}$$

★ $f_v = \eta A \frac{dv}{dx}$

$$\frac{f_v}{A} = \eta \cdot \frac{dv}{dx} = \eta \left(\frac{v - 0}{h} \right) = \text{shear stress}$$

★ $F_f = 6\pi \eta r v$

★ at terminal velocity $F_{\text{net}} = 0$



$$mg = B + 6\pi \eta r v$$

$$mg = B + 6\pi \eta r v$$

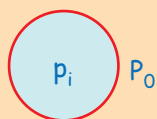
$$V = \frac{2}{9} \frac{gr^2}{\eta} (\rho_s - \rho_f)$$

$$\star S = F / \ell$$

$$SE = S.A$$

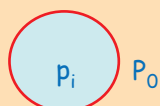
$$\star (WD)_{\text{ext}} = \Delta(S.E)$$

★



liquid drop

$$p_i - p_o = \frac{2S}{R}$$



soap bubble

$$p_i - p_o = \frac{4S}{R}$$

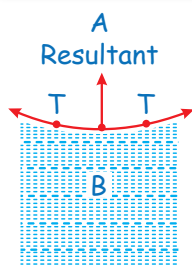
★ **Capillary tube**

$$h = \frac{2S}{R\rho g} = \frac{2S \cos \theta}{r\rho g}$$

$$\star r = \frac{R_1 R_2}{R_2 - R_1} \text{ (two soap bubble mix)}$$



★



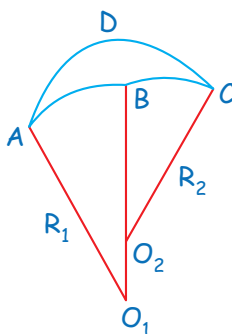
Concave Surface

$$p_A - p_B = \frac{2T}{r}$$

$$p_B - p_A = \frac{2T}{r}$$

Relation Between Surface Tension, Radii of Curvature and Excess Pressure on a Curved Surface

Let us consider a small element ABCD (figure) of a curved liquid surface which is convex on the upper side. R_1 and R_2 are the maximum and minimum radii of curvature respectively. Let p be the pressure on the concave side.



$$\text{Then, } p = T \left(\frac{1}{R_1} + \frac{1}{R_2} \right).$$

For Liquid Film

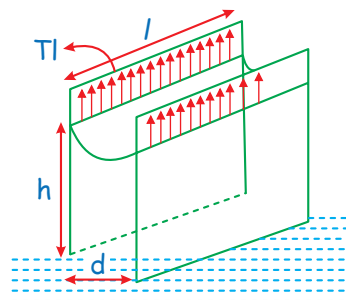
$$p = 2T \left(\frac{1}{R_1} + \frac{1}{R_2} \right),$$

(because a film has two surfaces.)

★ If two parallel plates with the spacing 'd' are placed in water, then height of rise

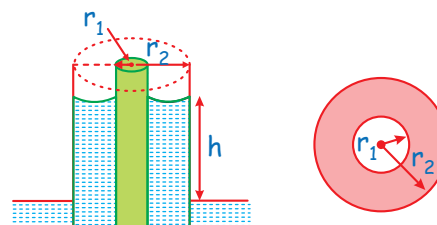
$$\Rightarrow 2Tl = \rho l h d g$$

$$\Rightarrow h = \frac{2T}{\rho d g} \quad (\theta = 0)$$



★ If two concentric tubes of radius ' r_1 ' and ' r_2 ' (inner one is solid) are placed in water reservoir, then height of rise

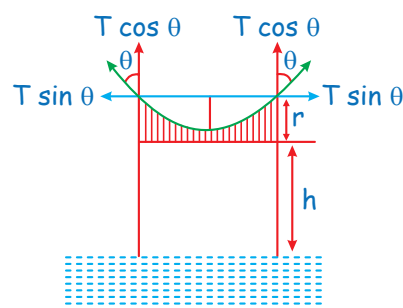
$$\Rightarrow T [2\pi r_1 + 2\pi r_2] = [\pi r_2^2 h - \pi r_1^2 h] \rho g$$



$$\Rightarrow h = \frac{2T}{(r_2 - r_1) \rho g} \quad (\theta = 0)$$

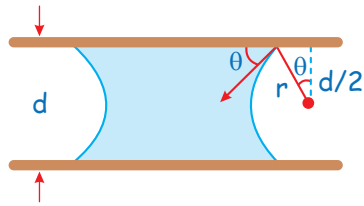
★ If weight of the liquid in the meniscus is also taken into account:

$$T \cos \theta \times 2\pi r = \left[\pi r^2 h + \frac{1}{3} \pi r^2 \times r \right] \rho g$$



$$\left[h + \frac{r}{3} \right] = \frac{2T \cos \theta}{r \rho g}$$

- ★ **Liquid between two plates:** When a small drop of water is placed between two glass plates put face to face, it forms a thin film which is concave outward along its boundary. Let 'R' and 'r' be the radii of curvature of the enclosed film in two perpendicular directions.



Hence, the pressure inside the film is less than the atmospheric pressure outside it by an amount p given by $p = T \left(\frac{1}{r} + \frac{1}{r = \infty} \right)$ and we have, $p = \frac{T}{r}$.

If d be the distance between the two plates and θ the angle of contact for water and glass, then

from the figure $\cos \theta = \frac{d}{2r}$ or $\frac{1}{r} = \frac{2 \cos \theta}{d}$.

Substituting for $\frac{1}{r}$ in, we get $p = \frac{2T}{d} \cos \theta$.

θ can be taken zero water and glass, i.e. $\cos \theta = 1$. Thus, the upper plate is pressure downward by the atmospheric pressure minus $\frac{2T}{d}$. Hence, the resultant downward pressure acting on the upper plate is $\frac{2T}{d}$. If A be the arc of the

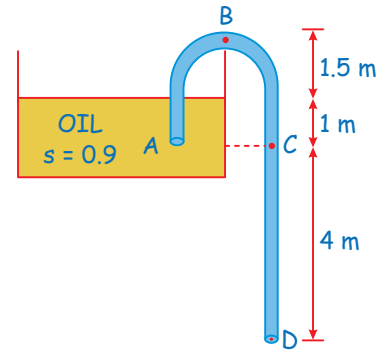
plate wetted by the film, the resultant force F pressing the upper plate downward is given by $F = \text{resultant pressure} \times \text{area} = \frac{2TA}{d}$. For very

nearly plane surface, d will be very small and hence the pressing force F very large. Therefore, it will be difficult to separate the two plates normally.

PRACTICE

- Q. A siphon tube is discharging a liquid of specific gravity 0.9 from a reservoir as shown in the figure.
- Find the velocity of the liquid through the siphon
 - Find the pressure at the highest point B.
 - Find the pressure at the points A (outside the tube) and C. State and explain the following:

- Would the rate of flow be more, less or the same if the liquid were water?
- Is there a limit on the maximum height of B above the liquid level in the reservoir?
- Is there a limit on the vertical depth of the right limb of the siphon? [Take $g = 10 \text{ m/s}^2$]



Sol. Assume datum at the free surface of the liquid.

- Applying Bernoulli's equation on point 1 and 2, as shown in the figure.

$$\frac{p_1}{\rho g} + \frac{v_1^2}{2g} + y_1 = \frac{p_2}{\rho g} + \frac{v_2^2}{2g} + y_2$$

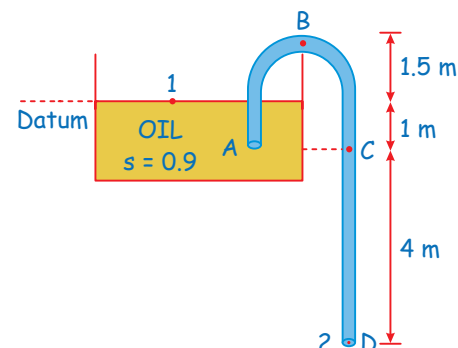
Here $p_1 = p_2 = p_0 = 10^5 \text{ N/m}^2$; $y_1 = 0$, $y_2 = -5 \text{ m}$

Since area of the tube is very small as compared to that of the reservoir, therefore,

$$v_1 \ll v_2. \text{ Thus } \frac{v_1^2}{2g} \approx 0$$

$$\therefore v_2 = \sqrt{2g(y_1 - y_2)} = \sqrt{2(10)(5)} = 10 \text{ m/s}$$

- Applying Bernoulli's equation at 1 and B.



$$\frac{p_B}{\rho g} + \frac{v_B^2}{2g} + y_B = \frac{p_1}{\rho g} + \frac{v_1^2}{2g} + y_1$$

$$\text{Here, } p_1 = 10^5 \text{ N/m}^2; \frac{v_1^2}{2g} \approx 0;$$

$$v_1 = 0, v_B = v_2 = 10 \text{ m/s}, y_B = 1.5 \text{ m}$$

$$\therefore p_B = p_1 - \frac{1}{2} \rho v_2^2 - \rho g y_B$$

$$\text{or } p_B = 10^5 - \frac{1}{2}(900)(10)^2 - (900)(10)(1.5) \\ = 41.5 \text{ kN/m}^2$$

- (c) Applying Bernoulli's equation at 1 and A

$$p_A = p_1 + \rho g (y_1 - y_A)$$

$$\text{or } p_A = 10^5 + (900)(10)(1) = 109 \text{ kN/m}^2.$$

Applying Bernoulli's equation at 1 and C,

$$p_C = p_1 - \frac{1}{2} \rho v_C^2 - \rho g y_C$$

$$= 10^5 - \frac{1}{2}(900)(10)^2 - (900)(10)(-1)$$

$$= 10^5 - 45000 + 9000 = 64 \text{ kN/m}^2.$$

- (d) The velocity of flow is independent of the density of the liquid, therefore, the discharge would remain the same.

- (e) Since the pressure at B is less than atmospheric, the liquid, therefore, has a tendency to get vapourised if the pressure becomes equal to the vapour pressure of it. Thus, $p_B > p_{\text{vapour}}$.

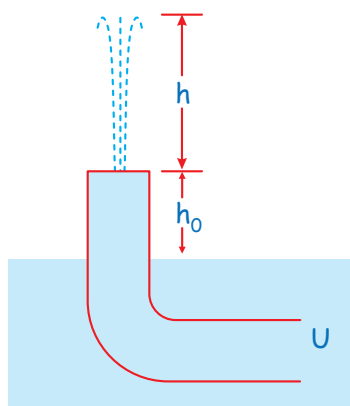
- (f) The velocity of flow depends on the depth of the point D, below the free surface

$$\frac{v^2}{2g} = y_1 - y_2 = H \text{ and}$$

$$p_B = p_1 - \frac{1}{2} \rho v^2 - \rho g y_B = p_1 - \rho g H - \rho g y_B$$

For working of siphon, $H \neq 0$, and H should not be high enough so that p_B may not reduce to vapour pressure.

- Q.** A bent tube is lowered into a water stream as shown in Fig. The velocity of the stream relative to the tube is equal to $v = 2.5 \text{ m/s}$. The closed upper end of the tube located at the height $h_0 = 12 \text{ cm}$ has a small orifice. To what height h will the water jet spurt?



- Sol.** Let the velocity of the water jet, near the orifice be v' , then applying Bernoulli's theorem,

$$\frac{1}{2} \rho v^2 = h_0 \rho g + \frac{1}{2} \rho v'^2; v' = \sqrt{v^2 - 2gh_0} \quad \dots(i)$$

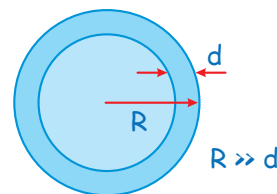
Here the pressure term on both sides is the same and equal to atmospheric pressure.

Now, if it rises upto a height then at this height, whole of its kinetic energy will be converted into potential energy. So,

$$\frac{1}{2} \rho v'^2 = \rho g h \text{ or } h = \frac{v'^2}{2g}$$

$$= \frac{v^2}{2g} - h_0 = 20 \text{ cm [Using equation (i)]}$$

- Q.** A soap bubble has radius R and thickness d ($d \ll R$) as shown. It collapse into a spherical drop. Then to energy released, and pressure in that process is



$$\text{Sol. } 2 \times 4\pi R^2 \times d = \frac{4}{3} \pi r^3$$

$$6R^2 d = r^3 \Rightarrow r = (6R^2 d)^{\frac{1}{3}}$$

$$S' = 4\pi r^2 = 4\pi (6R^2 d)^{\frac{2}{3}}$$

$$\Delta U = T \times (2 \times 4\pi R^2 - 4\pi r^2) \quad [\because \Delta U = T \times \Delta S]$$

$$= 4T\pi \left[2R^2 - (6R^2 d)^{\frac{2}{3}} \right].$$

- Q.** Pressure inside two soap bubbles are 1.06 atm and 1.08 atm. Ratio between their volumes is

$$\text{Sol. Excess pressure } (\Delta P_1) = P_1 - P_0 = 1.06 - 1 = 0.06 \text{ atm}$$

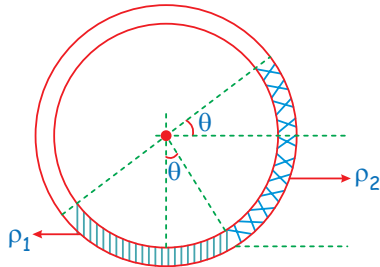
$$(\Delta P_2) = P_2 - P_0 = 1.08 - 1 = 0.08 \text{ atm}$$

$$\Delta P = \frac{4S}{R} \Rightarrow R = \frac{4S}{\Delta P} \Rightarrow \frac{R_1}{R_2} = \frac{\Delta P_2}{\Delta P_1}$$

$$\frac{R_1}{R_2} = \frac{0.08}{0.06} = \frac{8}{6} = \frac{4}{3}$$

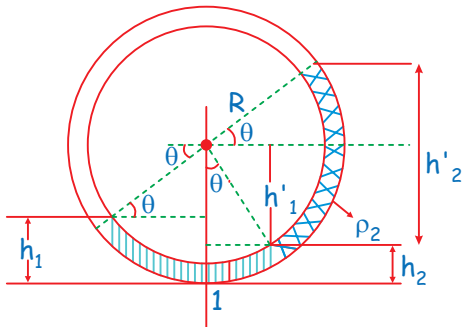
$$\therefore \frac{V_1}{V_2} = \left(\frac{R_1}{R_2} \right)^3 = \left(\frac{4}{3} \right)^3 = \frac{64}{27}$$

- Q.** A circular tube of uniform cross-section is filled with two liquids of densities ρ_1 and ρ_2 such that half of each liquid occupies a quarter of volume of the tube. If the line joining the free surface of the liquids makes an angle θ with horizontal, find the value of θ .



Sol. Let us find the pressure at the lowest point 1. Since, the liquid of density ρ_1 and height of liquid column h_1 is there, in right hand side of the point 1, we have.

$$P_1 = \rho_1 g h_1 \quad \dots (i)$$



Since two liquid columns of heights h_1 and h_2 and densities ρ_1 and ρ_2 are situated above the point 1, in left hand side, we have

$$P_2 = \rho_1 g h_2 + \rho_2 g h'_2 \quad \dots (ii)$$

Equating P_1 and P_2 from equation (i) and (ii), we have

$$\rho_1 h_2 + \rho_2 h'_2 = \rho_1 h_1 \quad \dots (iii)$$

Substituting $h'_2 = R \sin \theta + R \cos \theta$

$$h_2 = R(1 - \cos \theta)$$

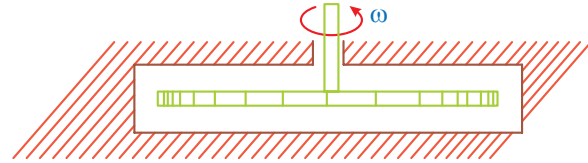
and $h_1 = R(1 - \sin \theta)$ in eqn. (iii), we have

$$\rho_1 R(1 - \cos \theta) + \rho_2 R(\sin \theta + \cos \theta) = \rho_1 R(1 - \sin \theta)$$

$$\text{This gives; } \frac{\cos \theta + \sin \theta}{\cos \theta - \sin \theta} = \frac{\rho_1}{\rho_2}$$

$$\tan \theta = \frac{\rho_1 - \rho_2}{\rho_1 + \rho_2}$$

- Q.** A thin horizontal disc of radius $R = 10$ cm is located within a cylindrical cavity filled with oil whose viscosity $\eta = 0.08$ P. The clearance between the disc and the horizontal planes



of the cavity is equal to $h = 1.0$ mm. Find the power developed by the viscous forces acting on the disc when it rotates with the angular velocity $\omega = 60$ rad/s. The end effects are to be neglected.

Sol. When the disc rotates, the fluid in contact with it corotates but the fluid in contact with the walls of the cavity does not rotate. A velocity gradient is then set up leading to viscous forces.

At a distance r from the axis the linear velocity is ωr so there is a velocity gradient

$$\frac{\omega r}{h} \text{ both in the upper and lower clearance.}$$

The corresponding force on the element whose radial width is dr is

$$F = \eta 2\pi r dr \frac{\omega r}{h} \text{ (from the formula } F = \eta A \frac{dv}{dx} \text{)}$$

The torque due to this force is

$$\tau = \eta 2\pi r dr \frac{\omega r}{h} r$$

and the net torque considering both the upper and lower clearance is

$$\tau_{\text{net}} = 2 \int_0^R \eta 2\pi r^3 dr \frac{\omega}{h} = \pi R^4 \omega \eta / h$$

So power developed is

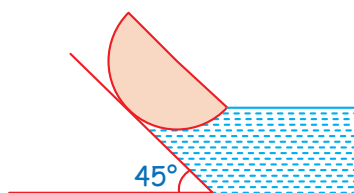
$$P = \pi R^4 \omega^2 \eta / h = 9.05 \text{ W (on putting the values).}$$

(As instructed end effects i.e. rotation of fluid in the clearance $r > R$ has been neglected.)

- Q.** A solid semicylinder of uniform density ρ rests in equilibrium on a rough inclined plane with a liquid of density ρ on its right as shown in Fig.

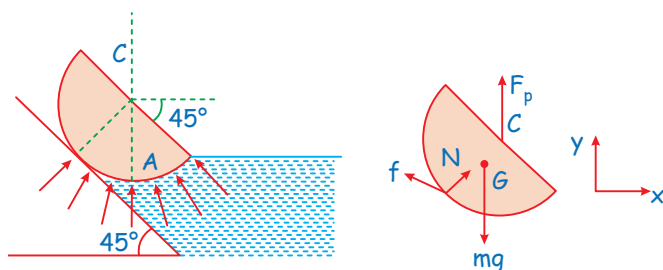
(a) Determine the minimum coefficient of friction to ensure equilibrium and

(b) The ratio ρ_1/ρ_2 . Assume that flat surface of the cylinder is parallel to the inclined plane.



Sol. The hydrostatic pressure force acts radially inwards, towards the axis of the cylinder that passes through and is normal to plane of paper. The torque due to this force about C is zero. Secondly the submerged portion is symmetrical about the line CA. Therefore

$$F_x = 0$$



From the diagram of pressure force it can be deduced

- (i) The torque of pressure force about C to zero
- (ii) The resultant pressure force acts vertically (by symmetry), i.e., $F_x = 0$.

Calculation of y-component of the pressure force,

$$\begin{aligned} F_y &= \int P dA \sin \theta \\ &= \int_{\pi/4}^{3\pi/4} \left[\rho g R \left(\sin \theta - \sin \frac{\pi}{4} \right) \right] (LR d\theta) \sin \theta \\ &= \rho g R^2 L \int_{\pi/4}^{3\pi/4} \left(\sin^2 \theta - \frac{1}{\sqrt{2}} \sin \theta \right) d\theta = \rho g R^2 L \left(\frac{\pi}{4} - \frac{1}{2} \right) \end{aligned}$$

... (i)

Balancing force in x-direction.

$$\frac{f}{\sqrt{2}} = \frac{N}{\sqrt{2}}$$

$$\text{or } f = N$$

$$\mu_s \geq \frac{f}{N}$$

Hence the minimum coefficient of static friction = 1.

$$\text{In y-direction, } F_p + \frac{1}{\sqrt{2}}(f + N) = mg$$

$$\left(\frac{\pi}{4} - \frac{1}{2} \right) LR^2 \rho g + \sqrt{2} f = mg \quad [\text{using (i)}] \quad \dots (ii)$$

Balancing torque about C,

$$fR = mg \frac{4}{3\pi} R \sin \frac{\pi}{4}, \quad f = \frac{4mg}{3\sqrt{2}\pi}$$

Putting in (ii),

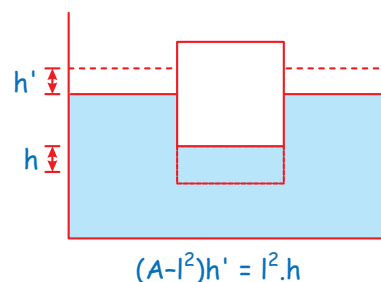
$$\text{or } \left(\frac{\pi}{4} - \frac{1}{2} \right) LR^2 \rho g + \frac{4mg}{3\pi} = mg$$

$$\frac{m}{LR^2 \rho} = \frac{3\pi(\pi - 2)}{4(3\pi - 4)}$$

$$\text{The required ratio, } \frac{\rho_1}{\rho_2} = \frac{2m}{L\rho\pi R^2} = \frac{3(\pi - 2)}{4(3\pi - 4)}$$

Q. A cubical block of wood has density $\rho_1 = 500 \text{ kg/m}^3$ and side $l = 30 \text{ cm}$. It is floating in a rectangular tank partially filled with water of density $\rho_2 = 1000 \text{ kg/m}^3$ and having base area, $A = 45 \text{ cm} \times 60 \text{ cm}$. Calculate work done to press the block slowly so that it is just immersed in water. ($g = 10 \text{ m/s}^2$)

Sol. $\frac{\rho_1}{\rho_2} = \frac{1}{2}$ i.e., in equilibrium, block is half immersed in water. Let h' be the increase in level when the block is pressed by an amount h . Then



Substituting the values, we get $h' = 0.5 h$. We have to immerse further $l/2$.

$$\text{Hence } \frac{l}{2} = h + h' = 1.5h \text{ or } h = \frac{l}{3}$$

When the block is depressed by h , extra upthrust

$$F = (h + h') \rho_2 l^2 \cdot g = 1.5 l^2 \rho_2 g h$$

Work has to be done against this upthrust.

$$\text{Hence } W = \int_0^{l/3} F dh = \int_0^{l/3} (1.5 l^2 \rho_2 g h) dh$$

Substituting the values, we get $W = 6.75 \text{ J}$